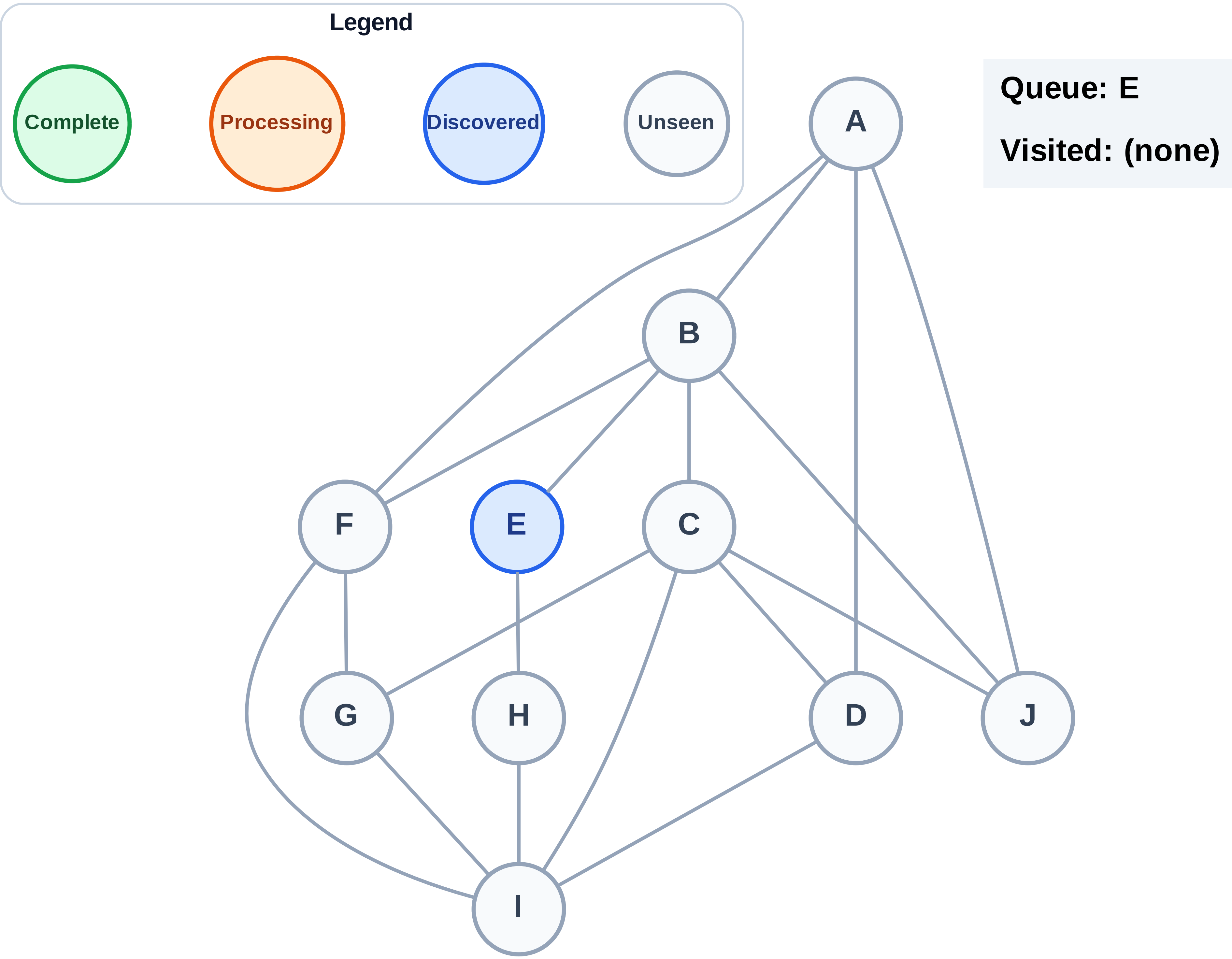
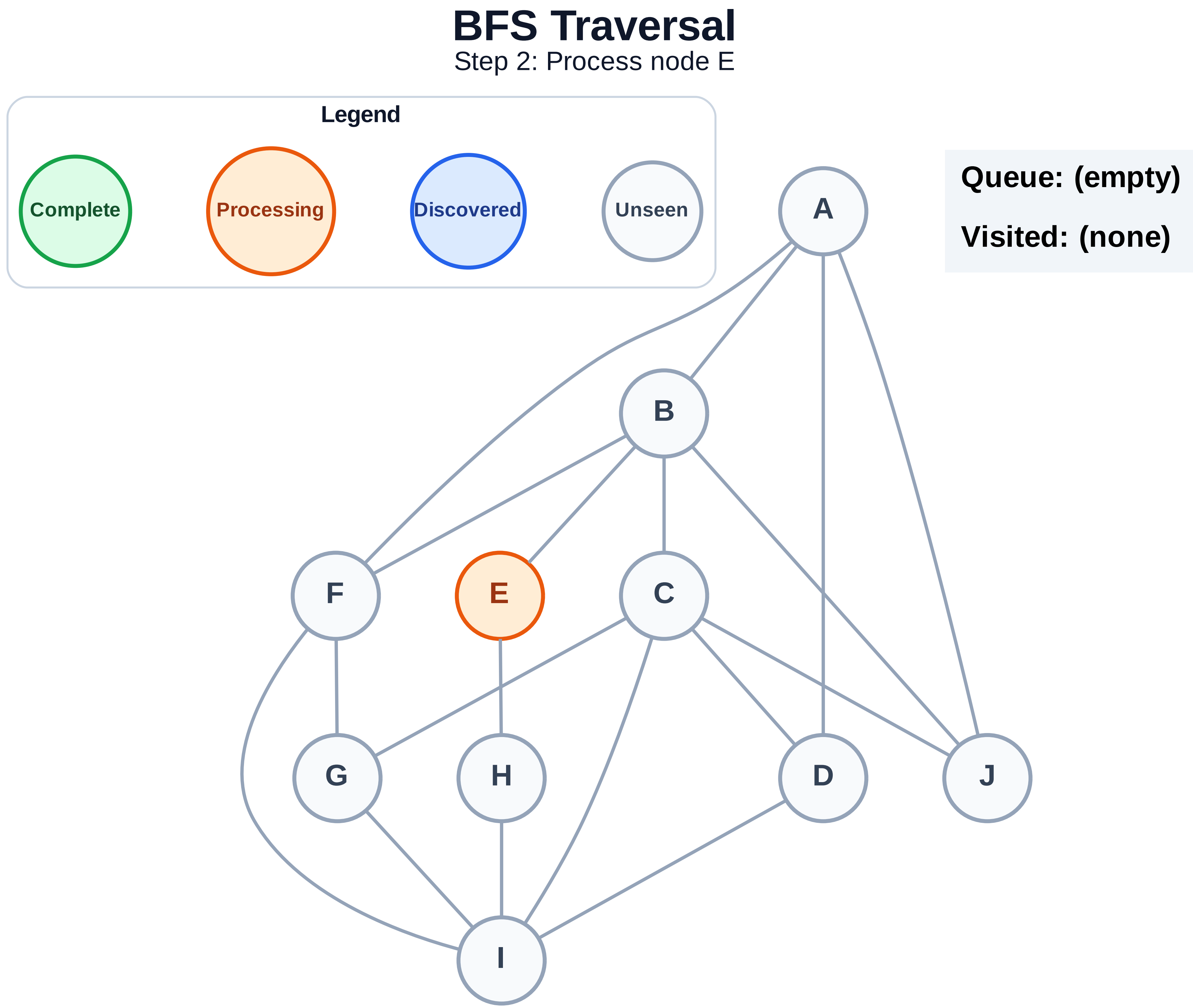
BFS Traversal

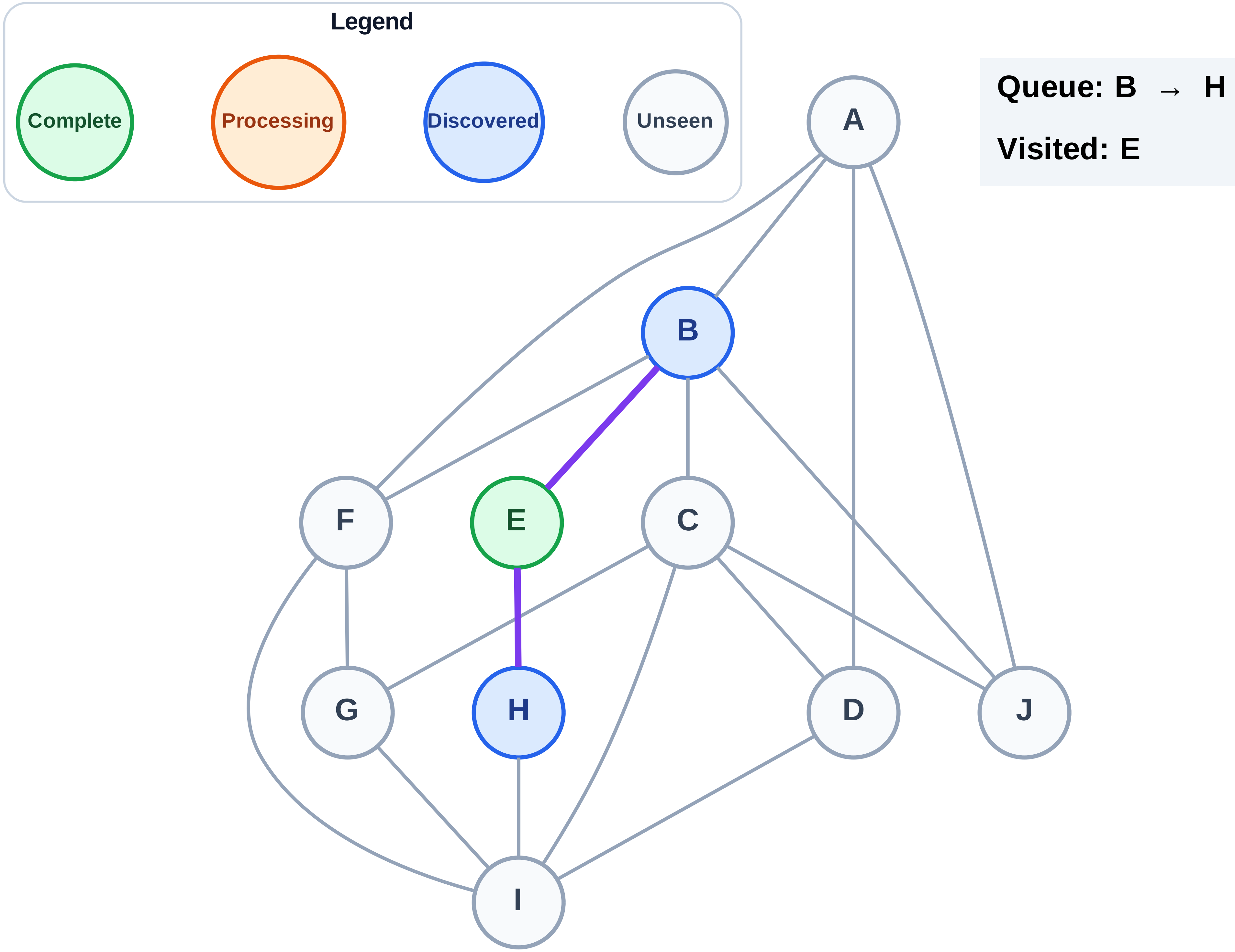
Step 1: Start at E




BFS Traversal
Step 2: Process node E
Legend
Complete
Processing
Discovered
Unseen
Queue: (empty)
Visited: (none)
A
B
C
D
E
F
G
H
I
J

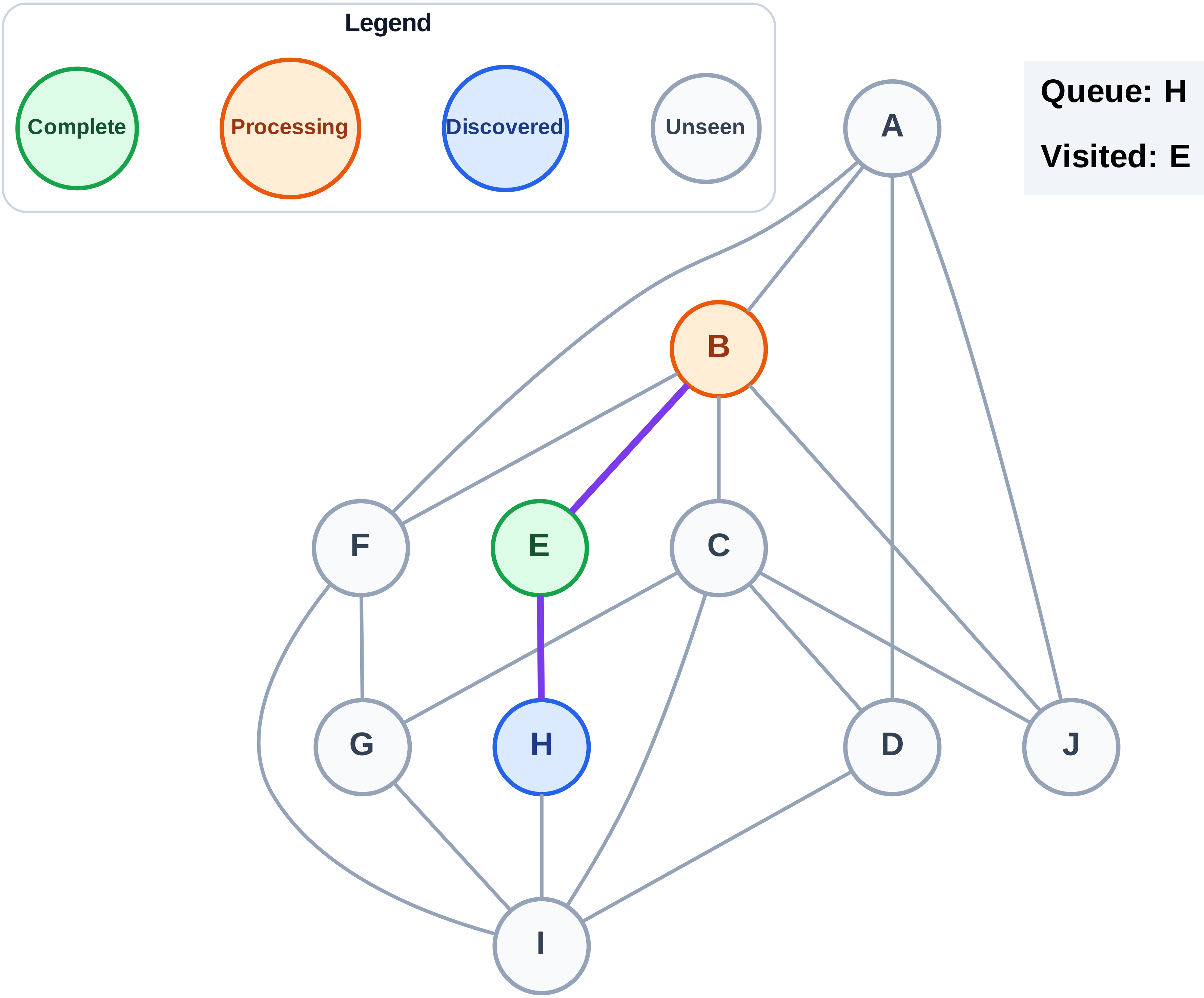
BFS Traversal

Step 3: Discover B, H from E



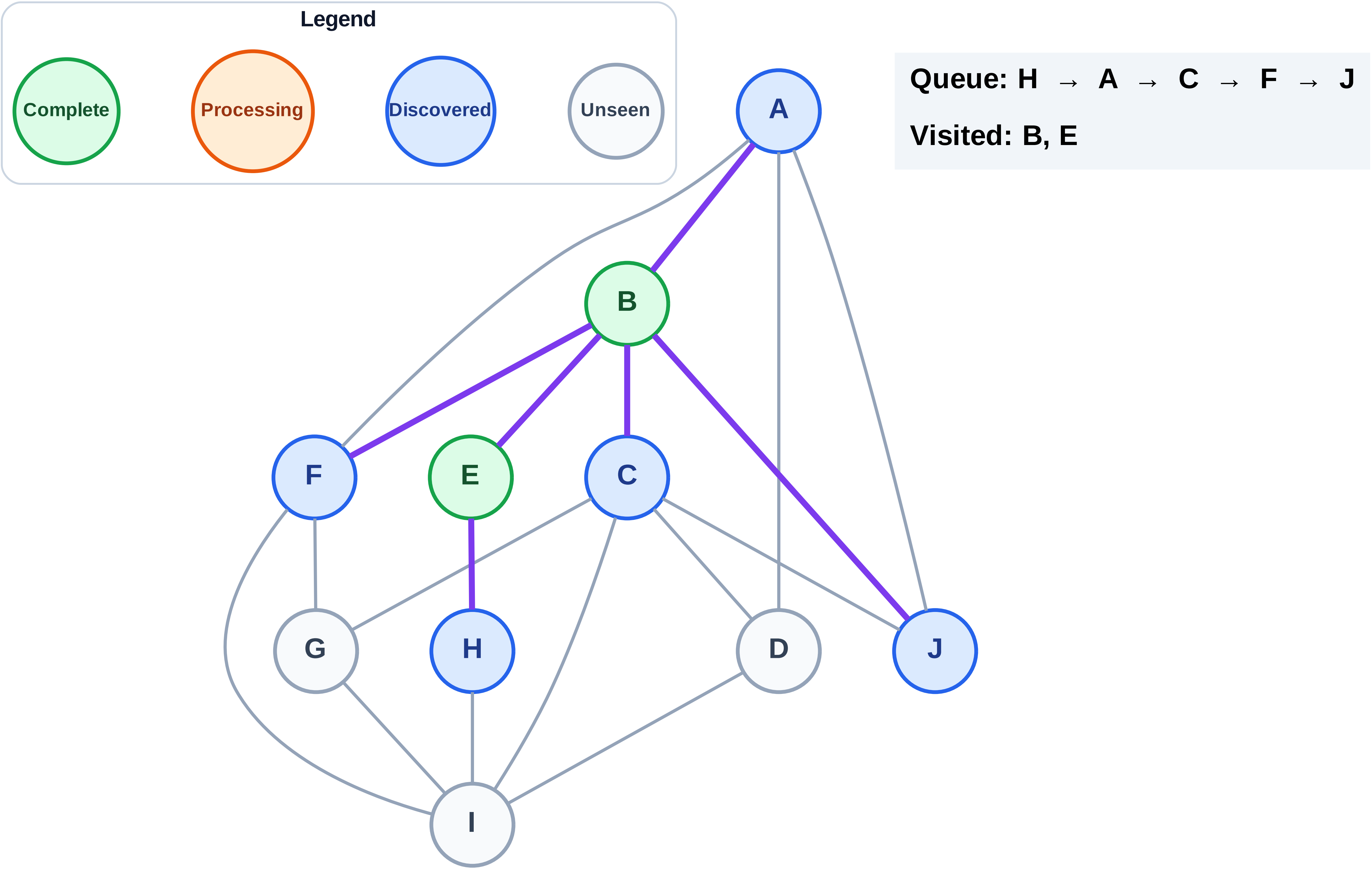
BFS Traversal

Step 4: Process node B



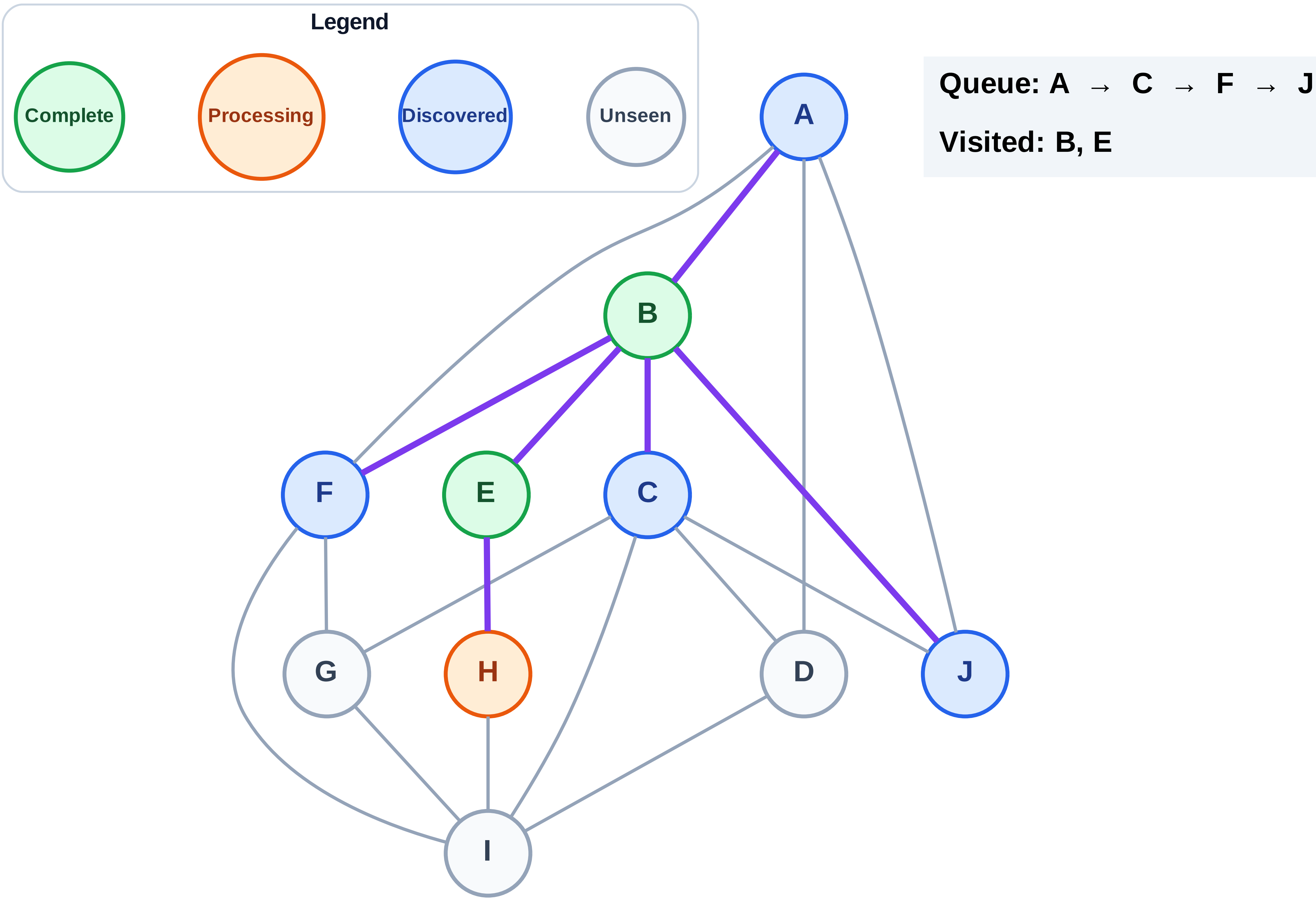
BFS Traversal

Step 5: Discover A, C, F, J from B



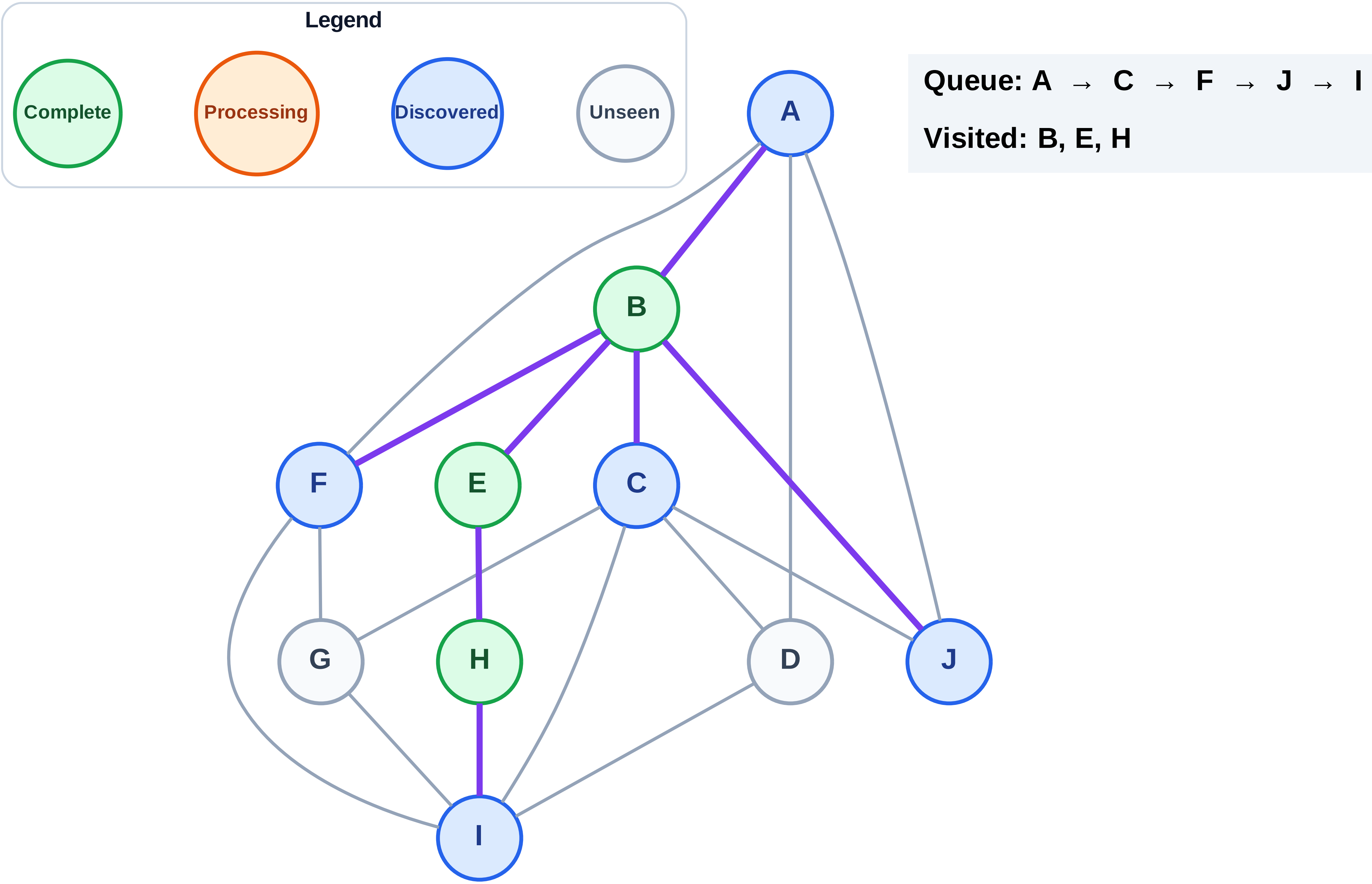
BFS Traversal

Step 6: Process node H



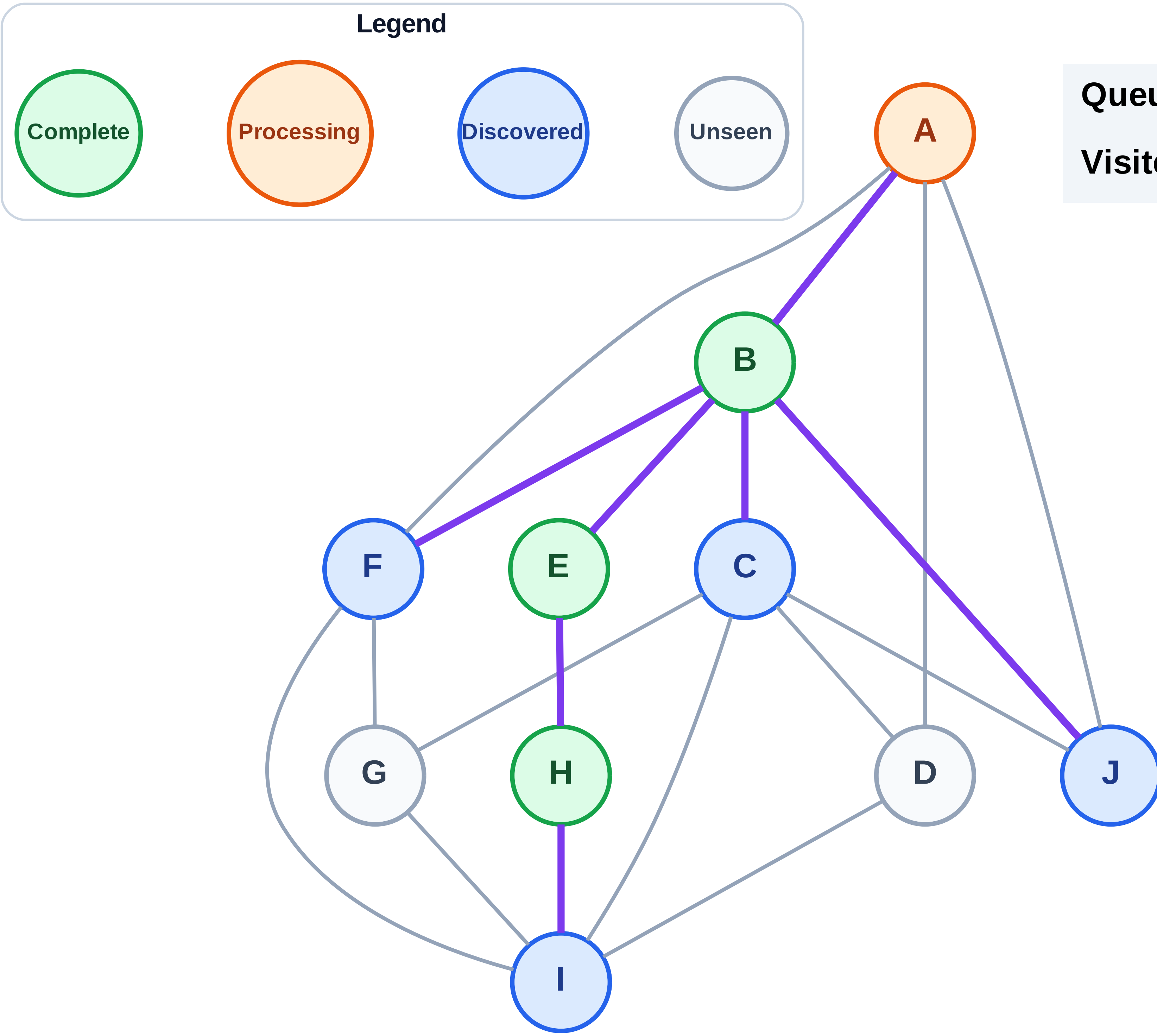
BFS Traversal

Step 7: Discover I from H



BFS Traversal

Step 8: Process node A

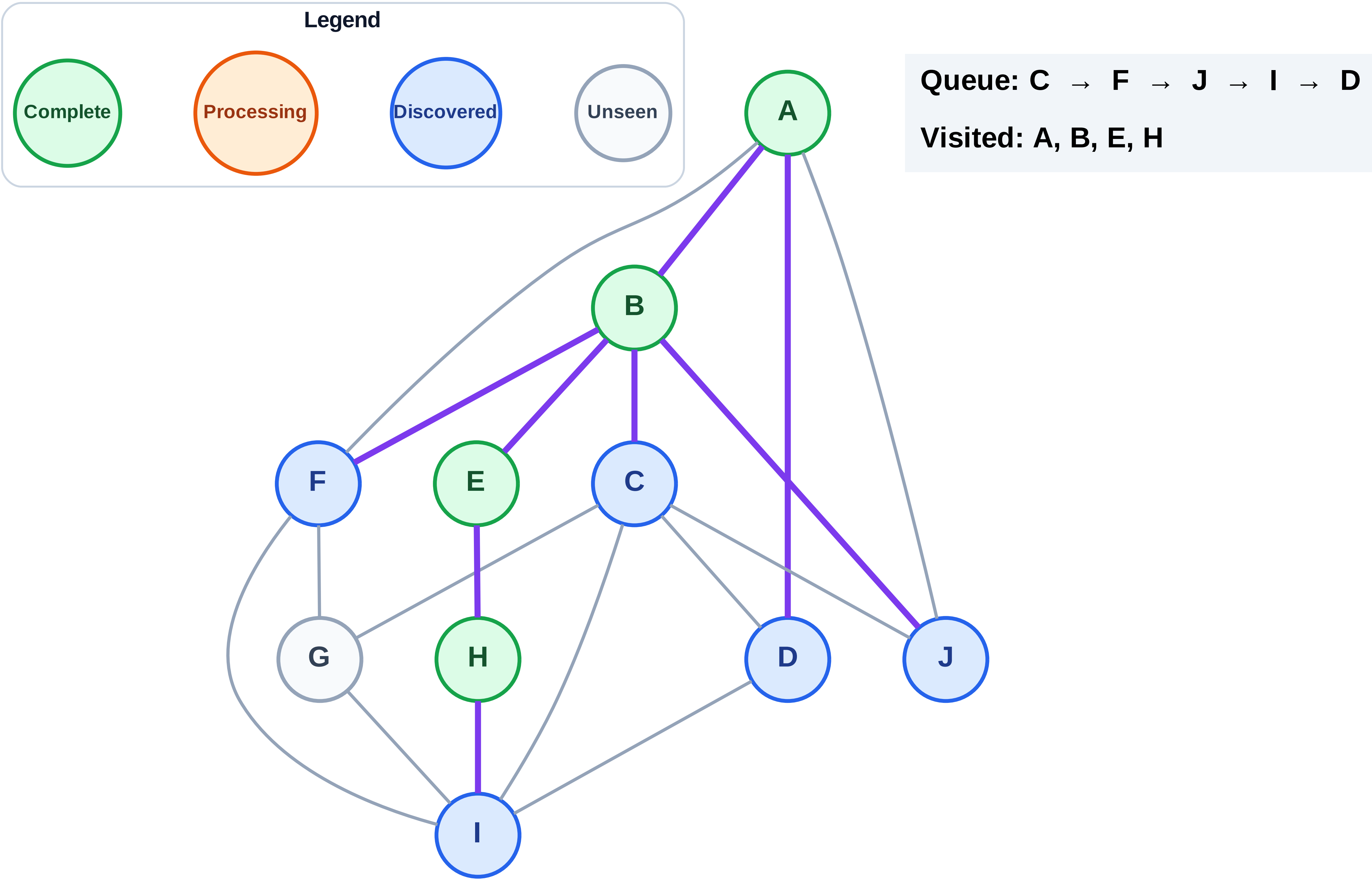


Queue: C → F → J → I

Visited: B, E, H

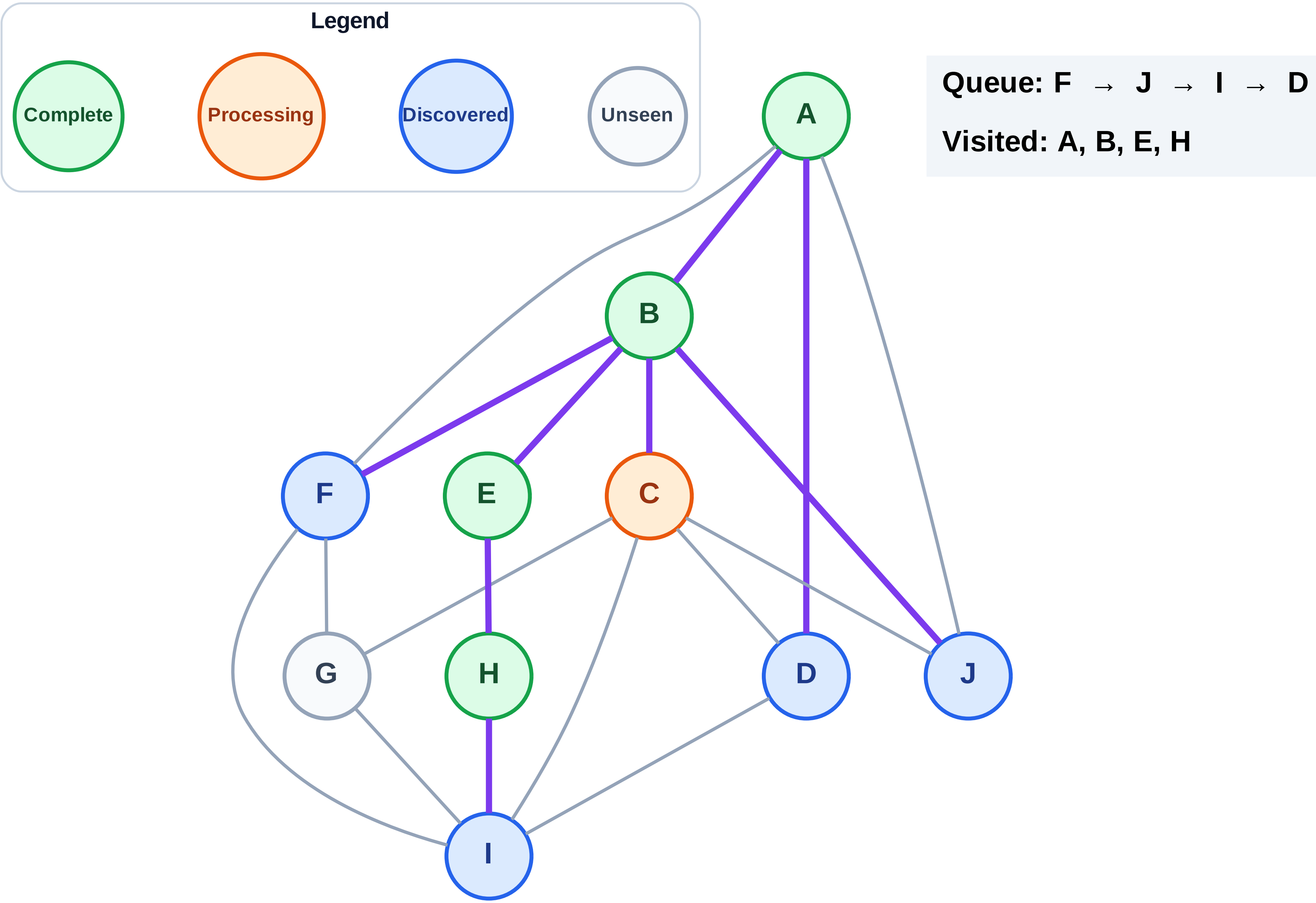
BFS Traversal

Step 9: Discover D from A



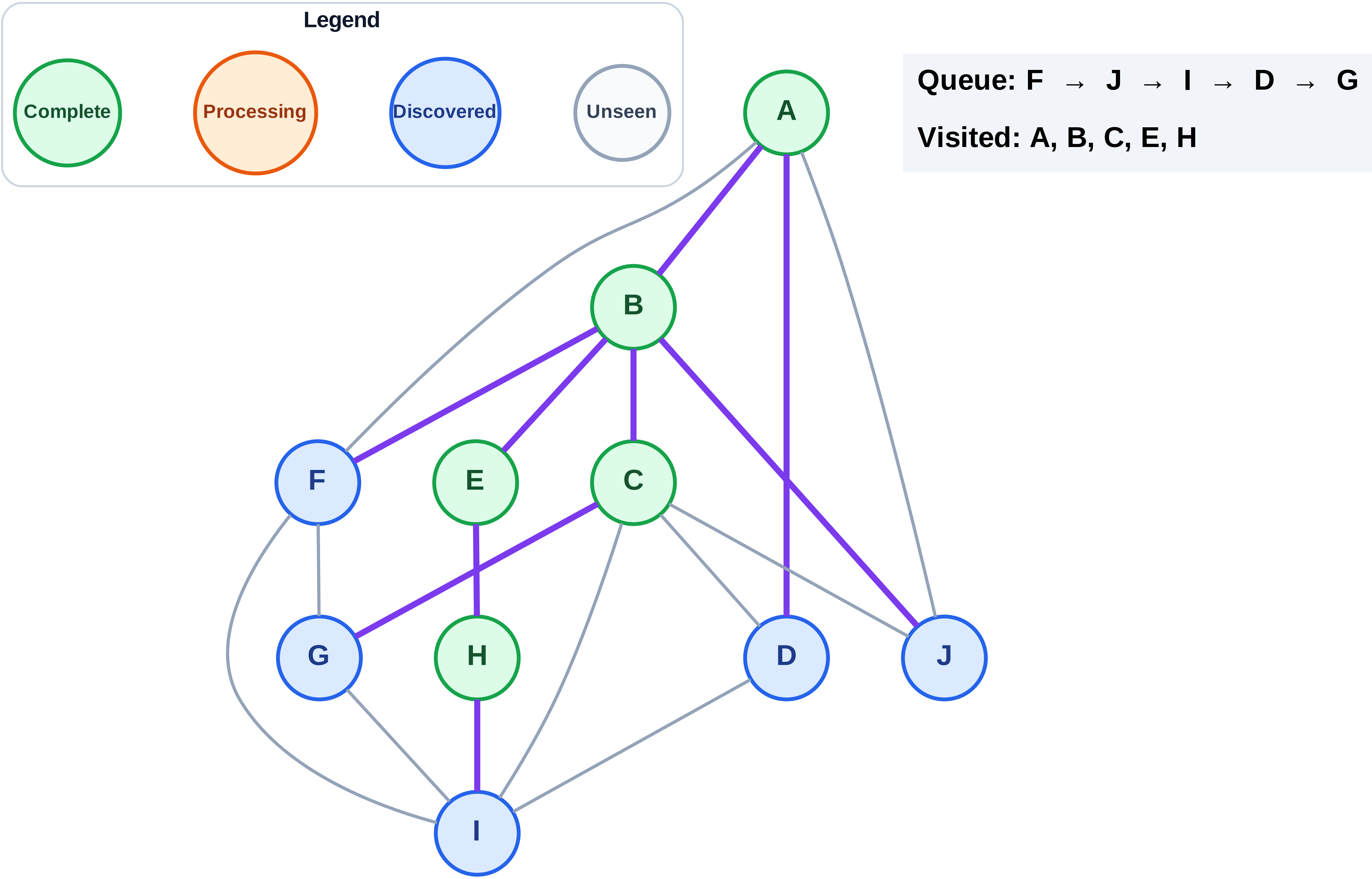
BFS Traversal

Step 10: Process node C



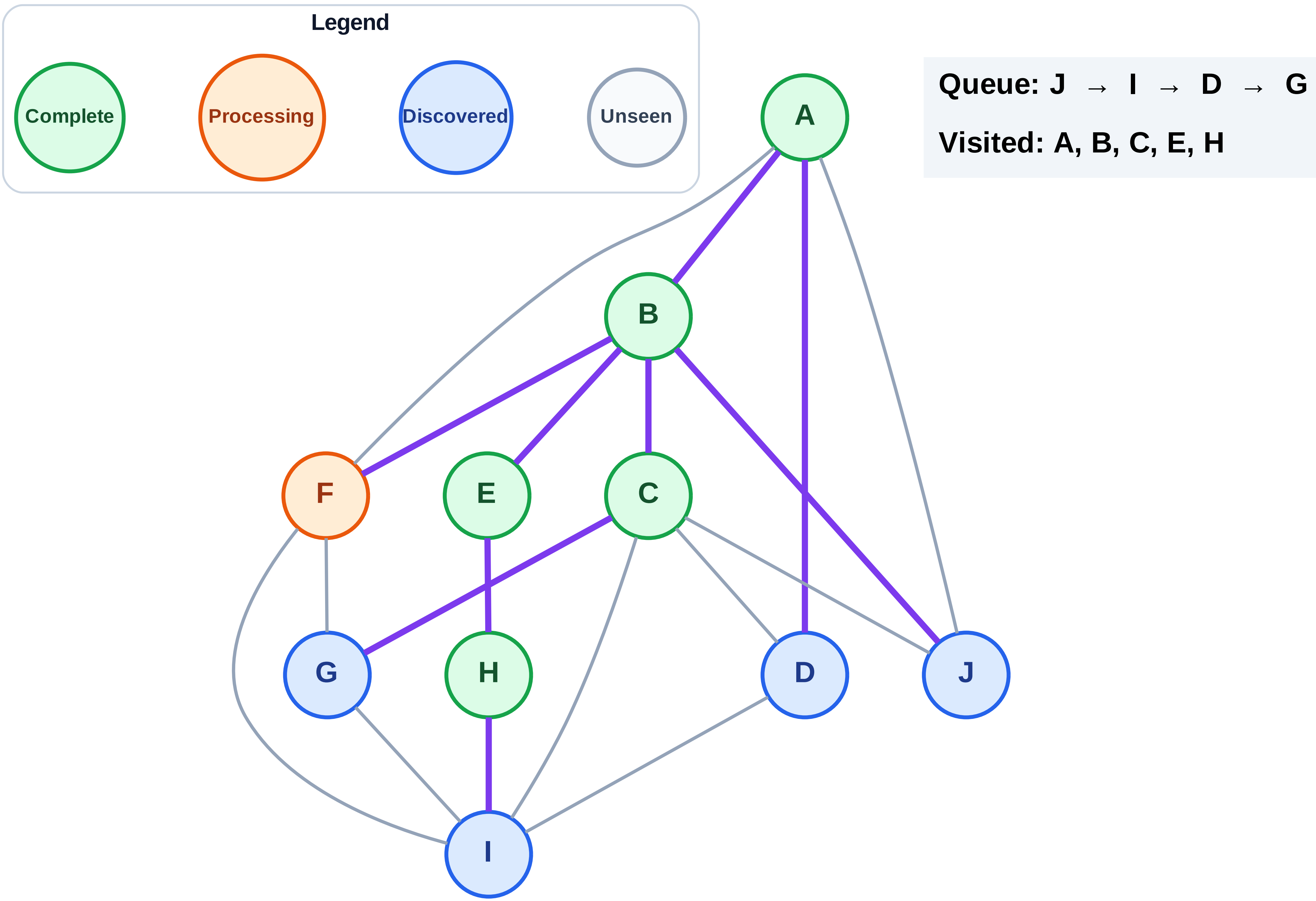
BFS Traversal

Step 11: Discover G from C



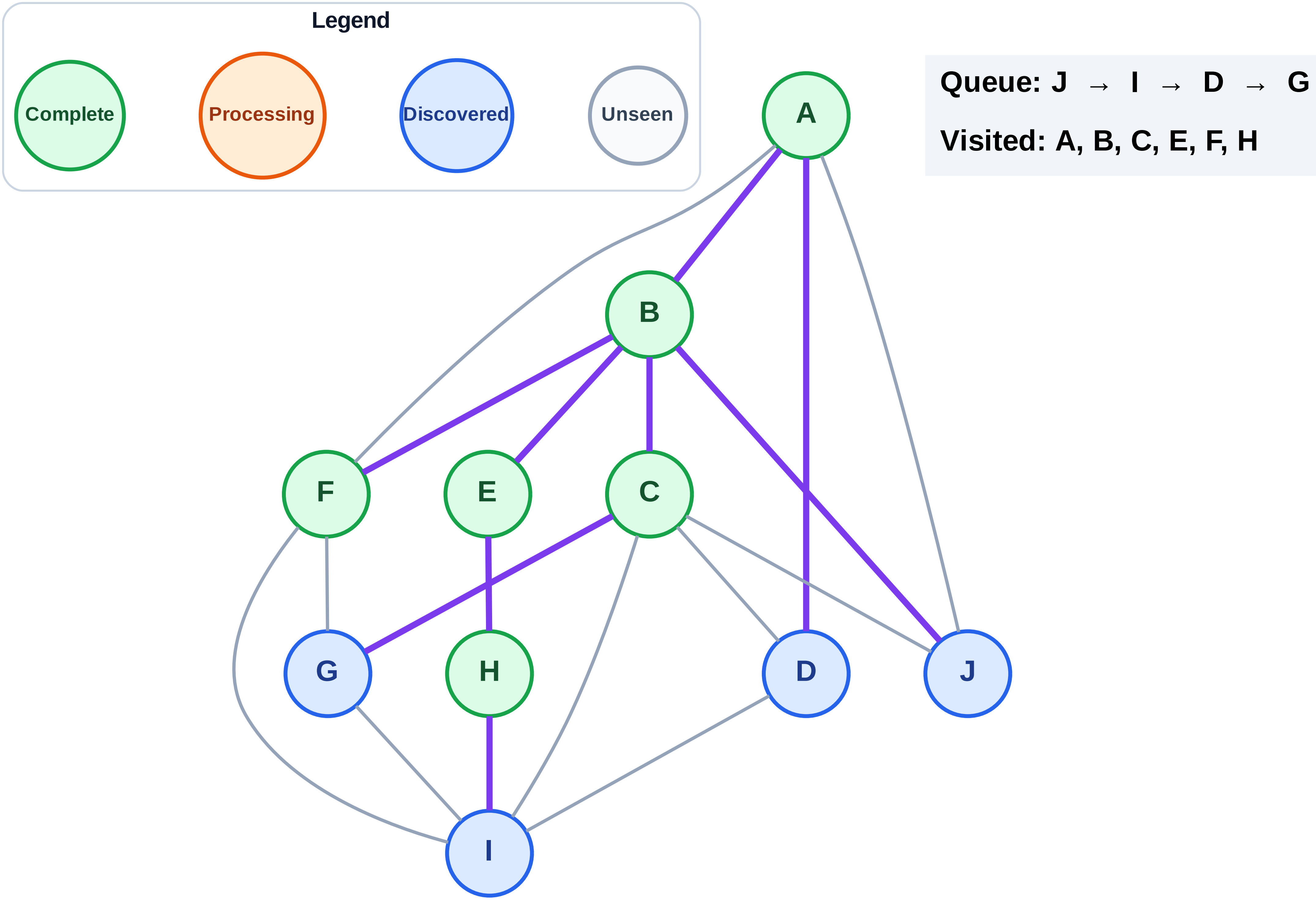
BFS Traversal

Step 12: Process node F



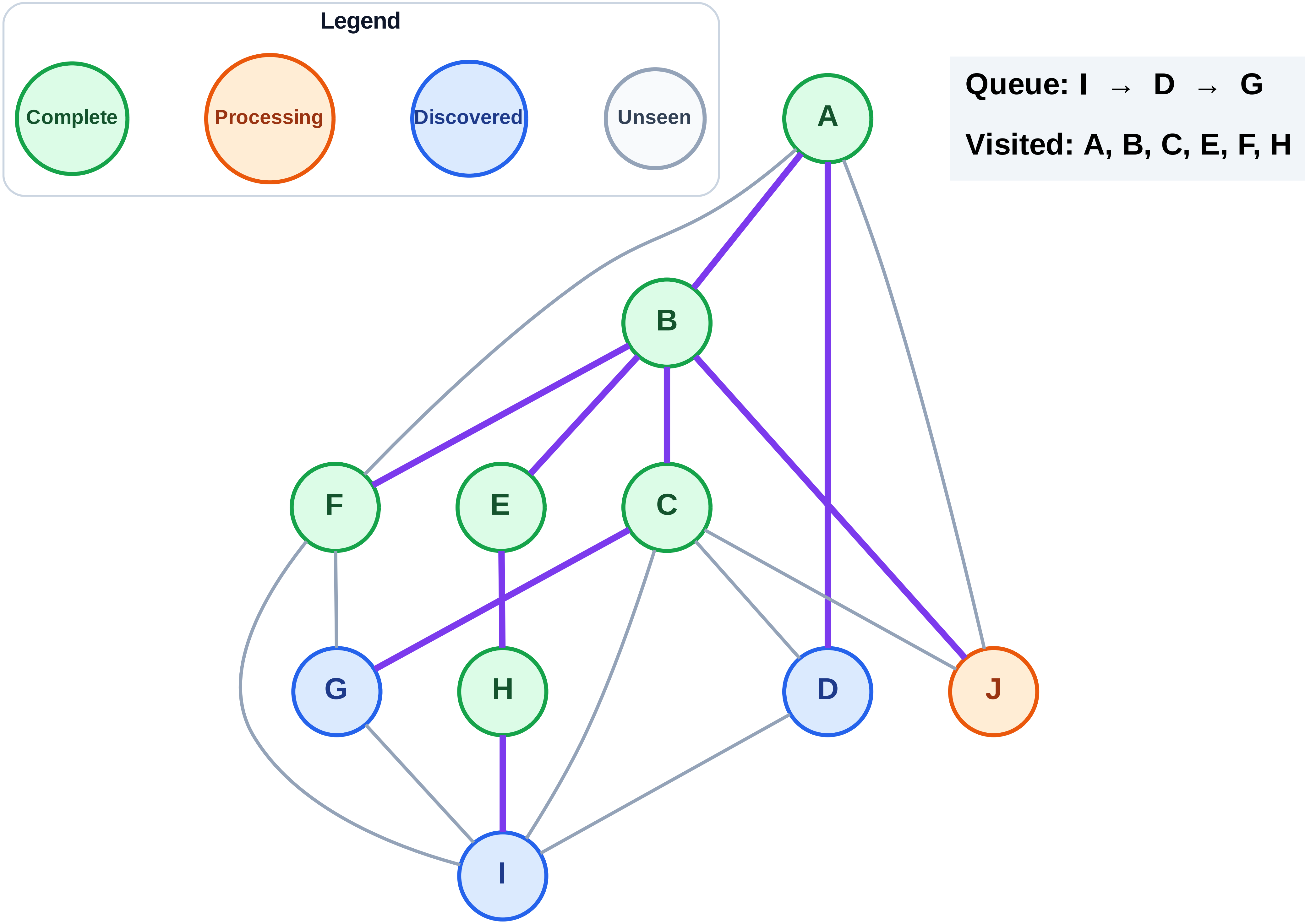
BFS Traversal

Step 13: No new neighbors from F



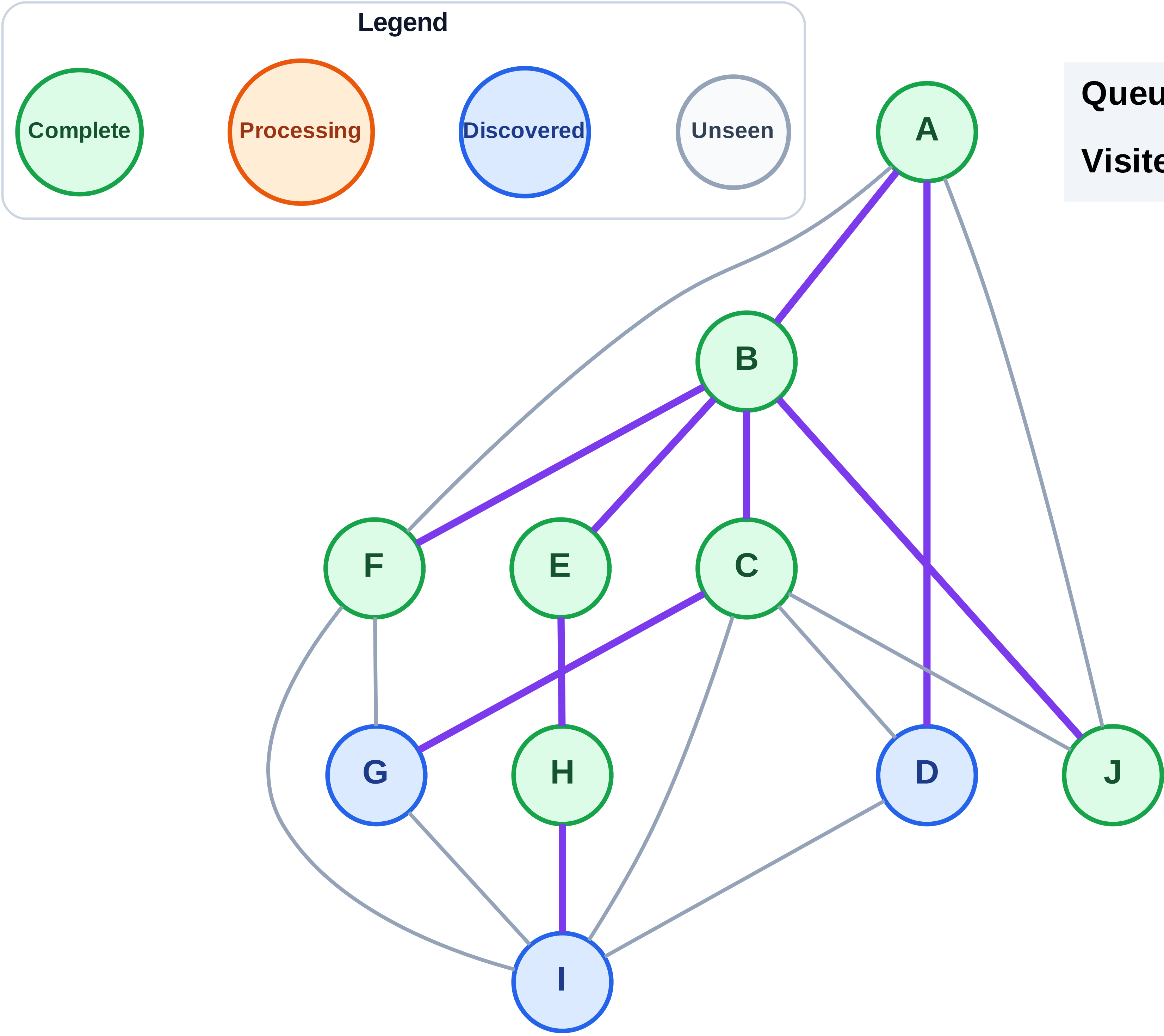
BFS Traversal

Step 14: Process node J



BFS Traversal

Step 15: No new neighbors from J

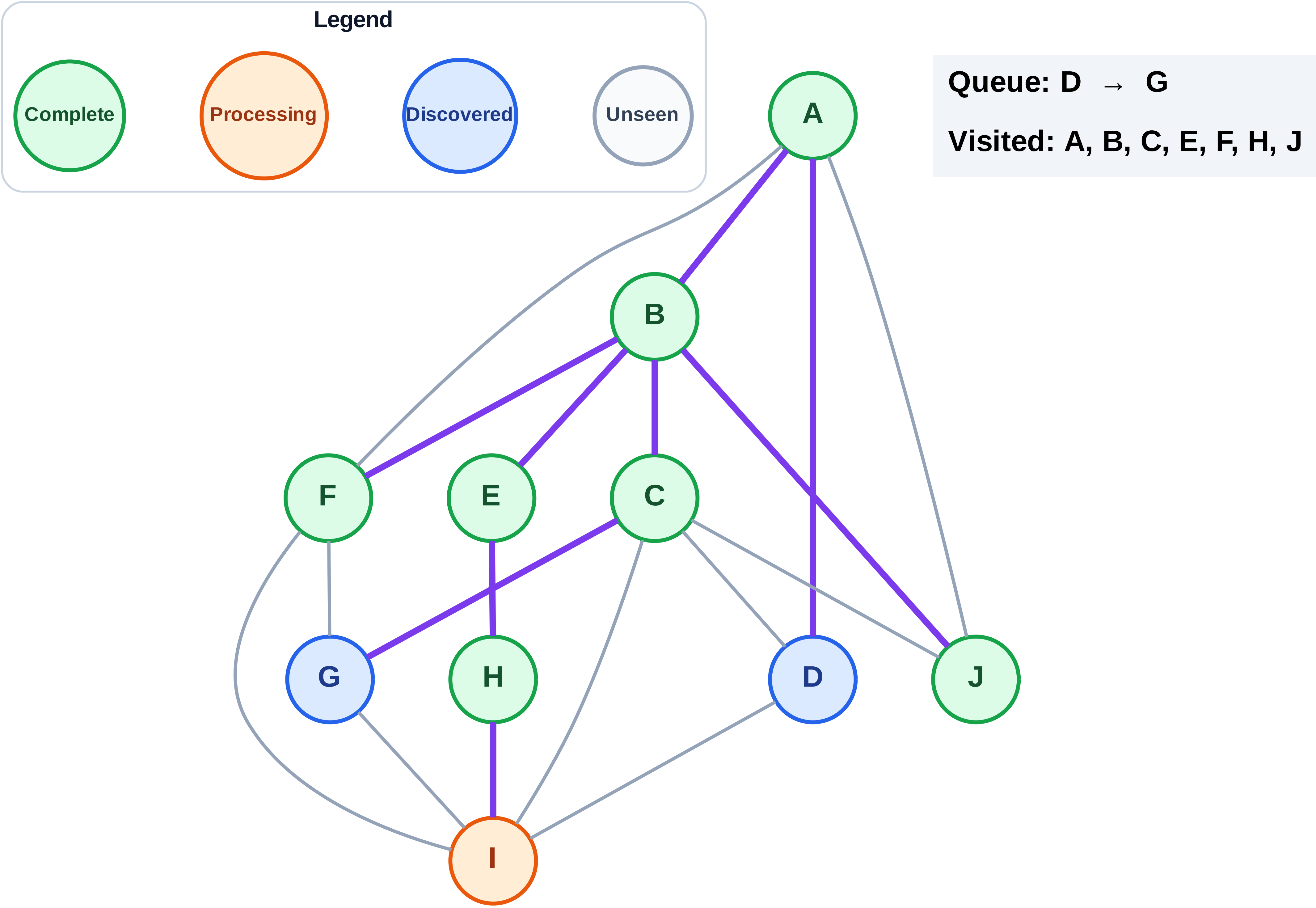


Queue: I → D → G

Visited: A, B, C, E, F, H, J

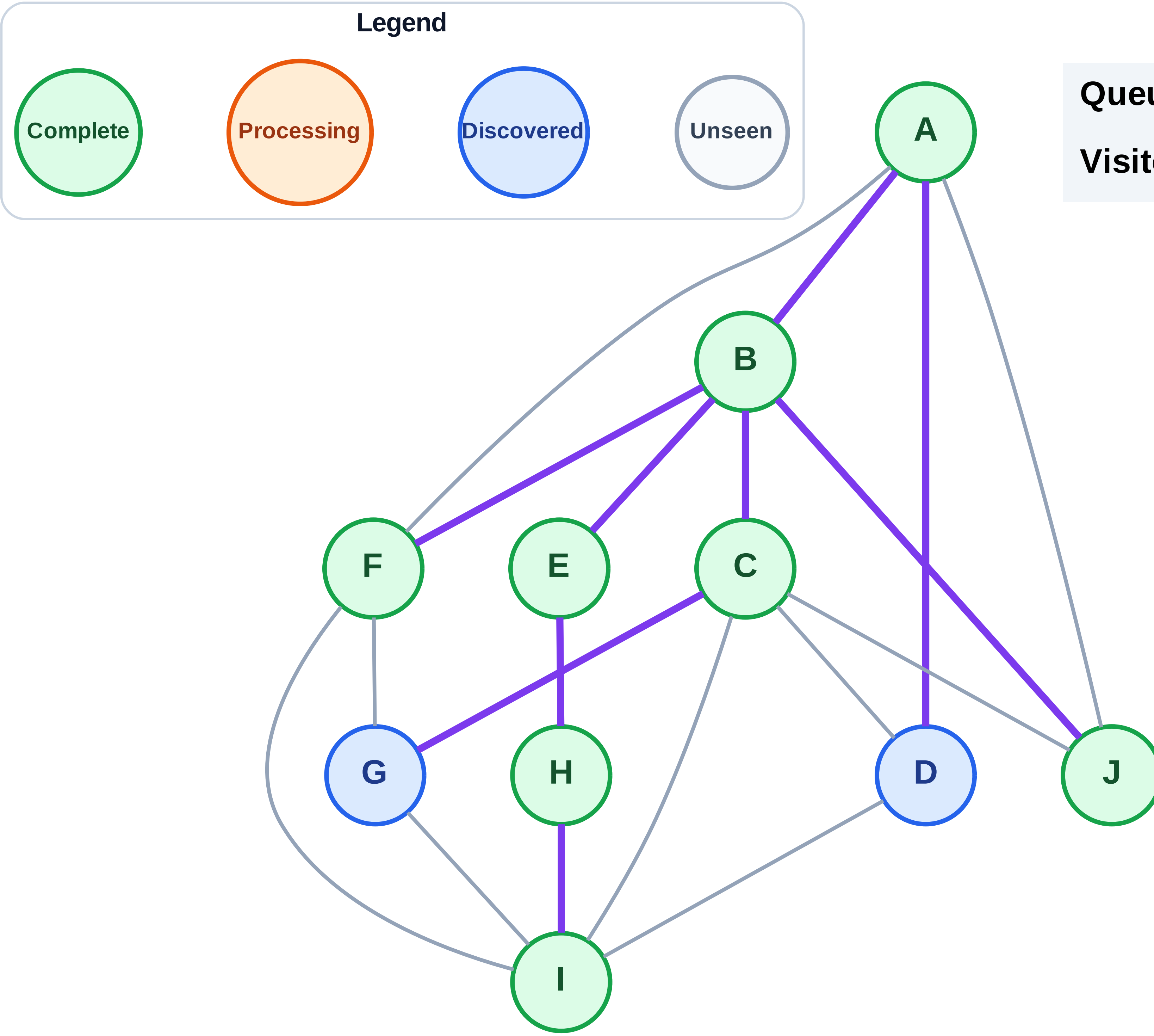
BFS Traversal

Step 16: Process node I



BFS Traversal

Step 17: No new neighbors from I

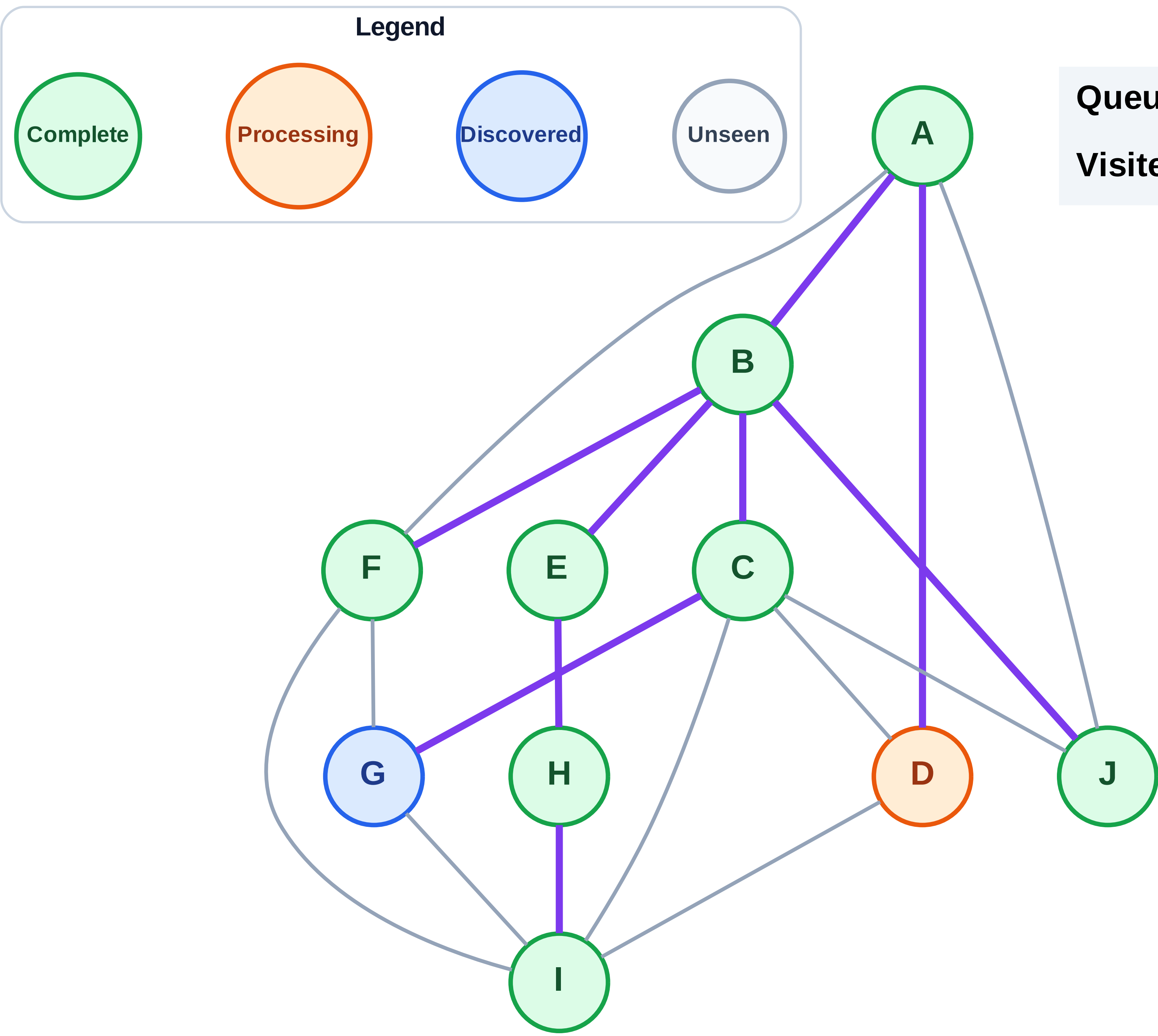


Queue: D → G

Visited: A, B, C, E, F, H, I, J

BFS Traversal

Step 18: Process node D

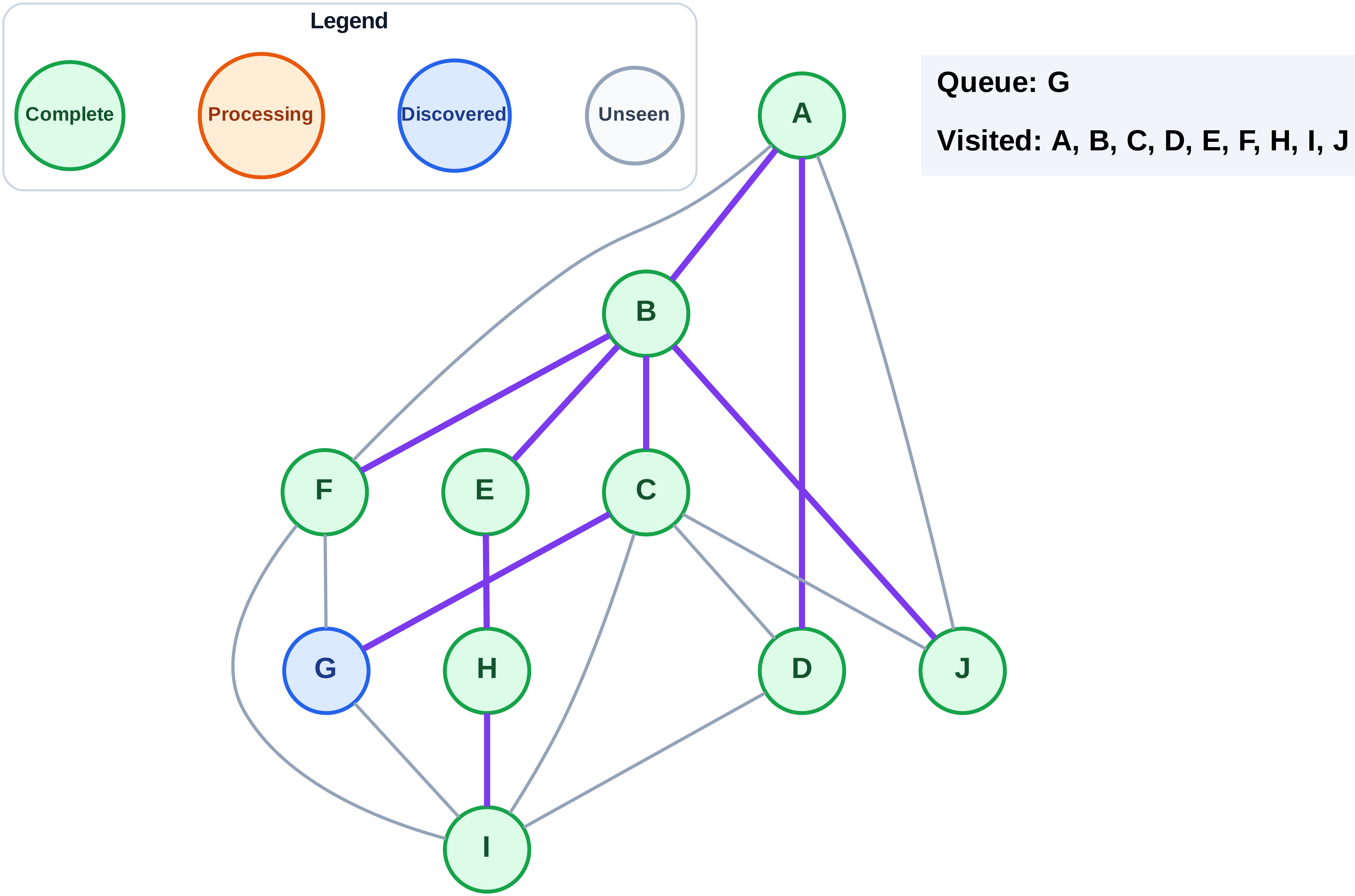


Queue: G

Visited: A, B, C, E, F, H, I, J

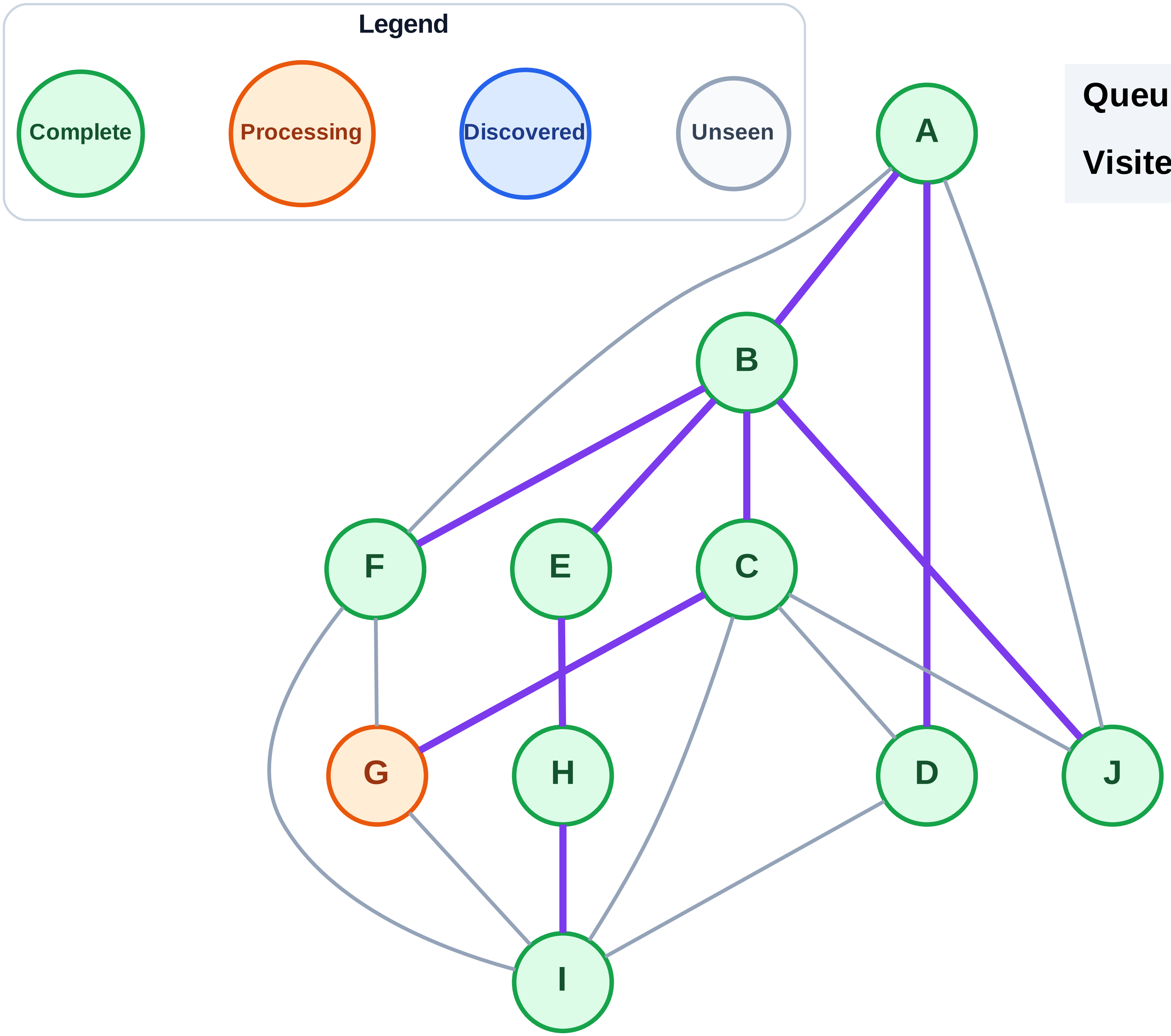
BFS Traversal

Step 19: No new neighbors from D



BFS Traversal

Step 20: Process node G

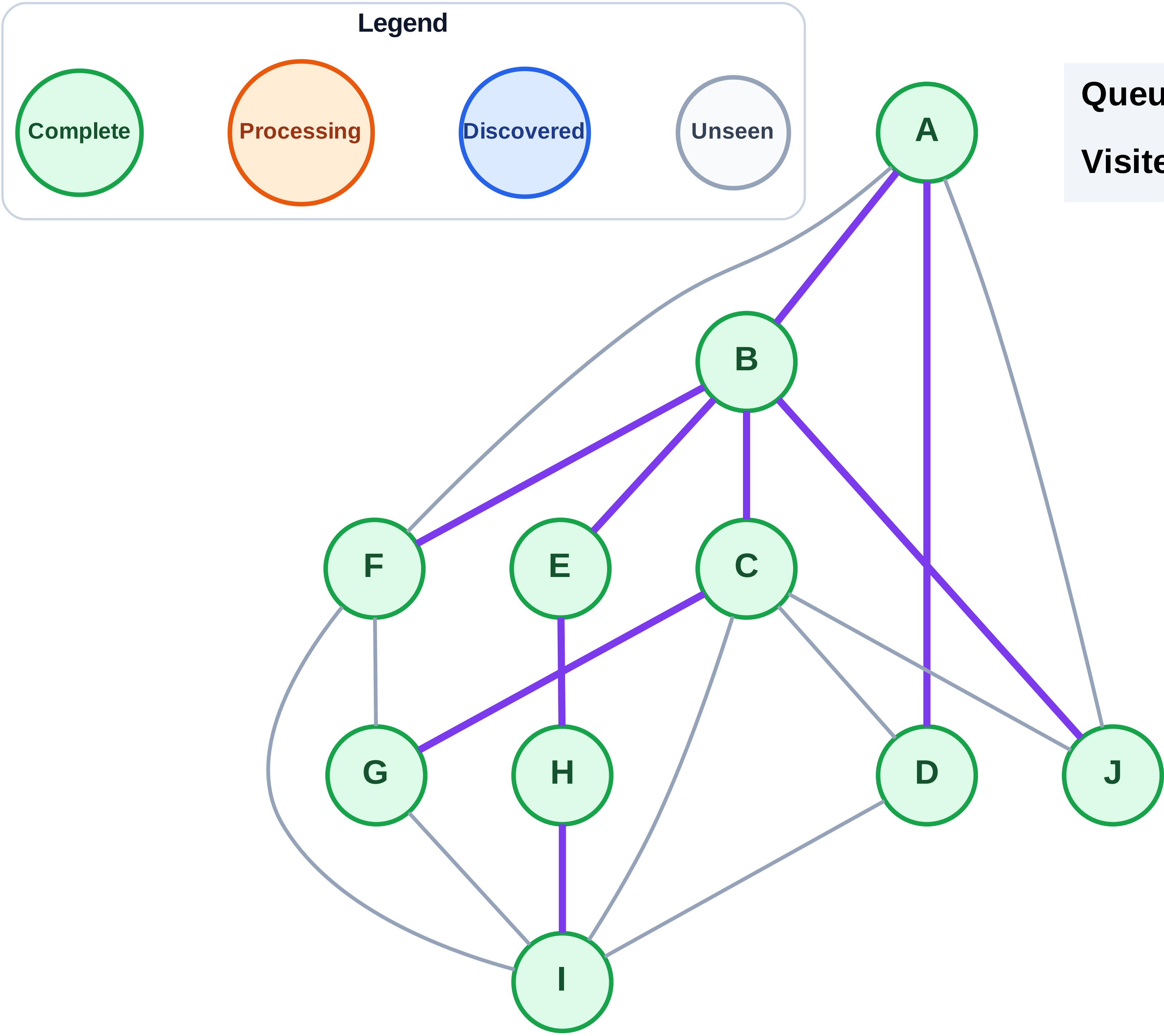


Queue: (empty)

Visited: A, B, C, D, E, F, H, I, J

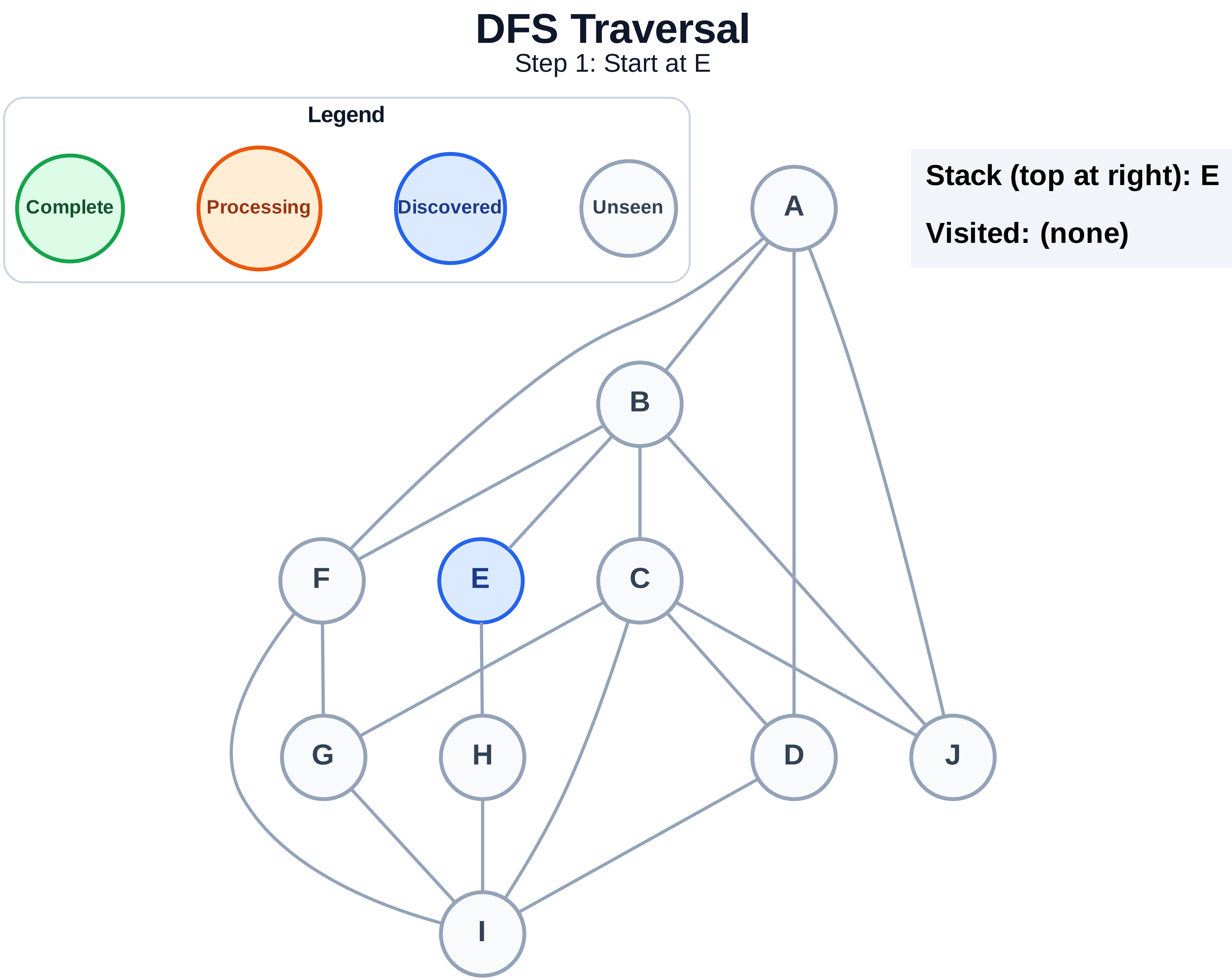
BFS Traversal

Step 21: No new neighbors from G



Queue: (empty)

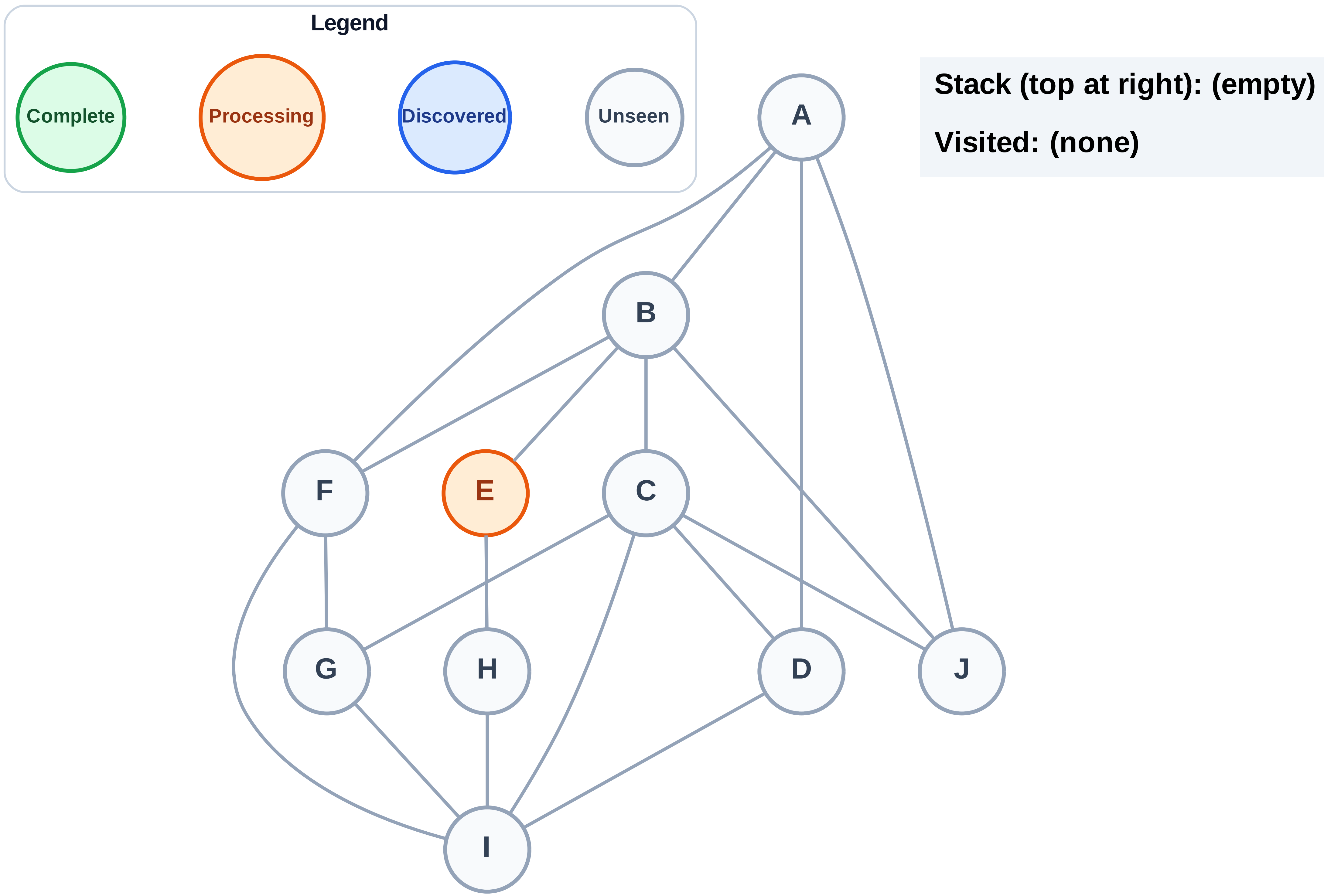
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal
Step 1: Start at E
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): E
Visited: (none)
A
B
F
E
C
G
H
D
J
I

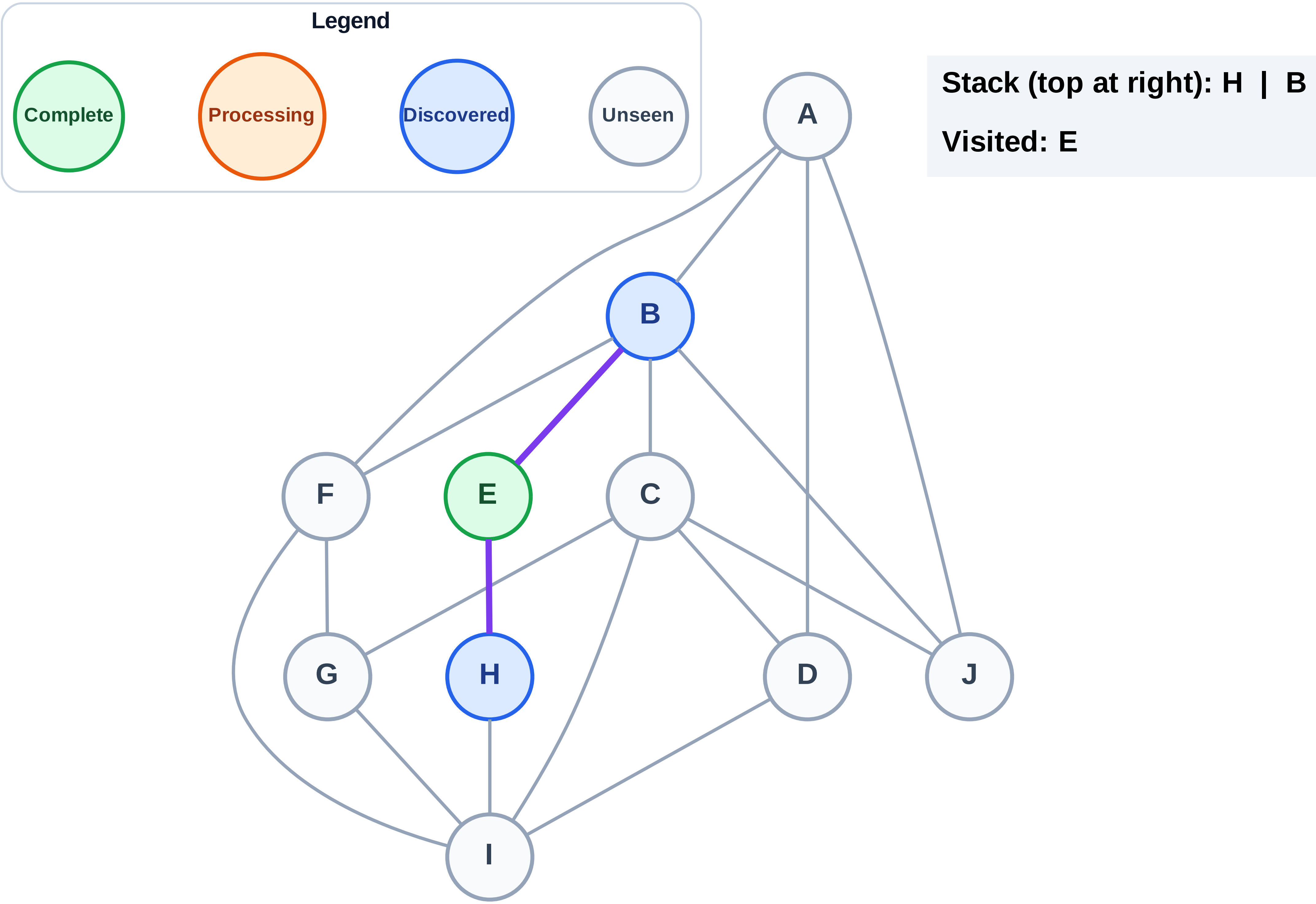
DFS Traversal

Step 2: Process node E



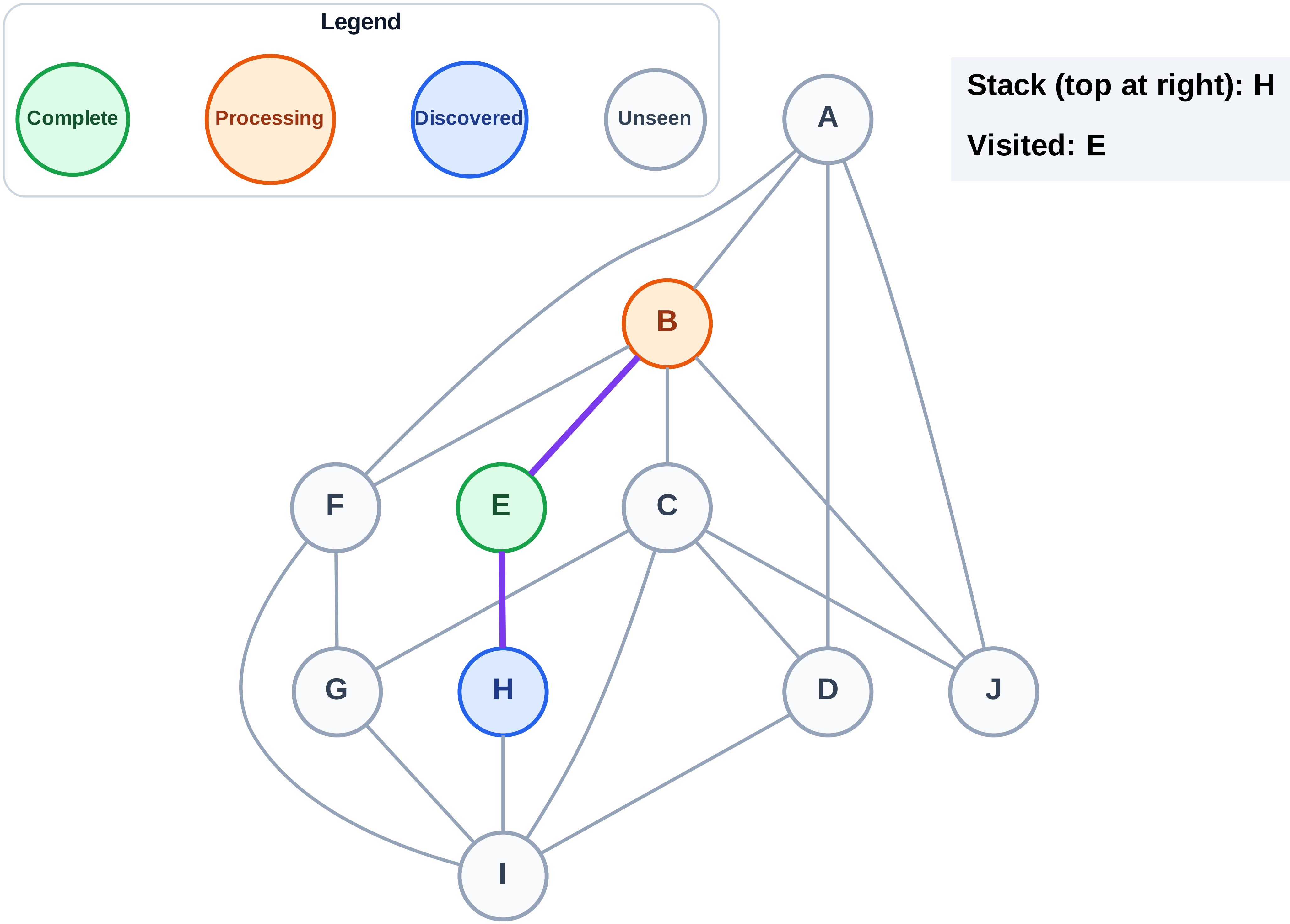
DFS Traversal

Step 3: Discover H, B from E



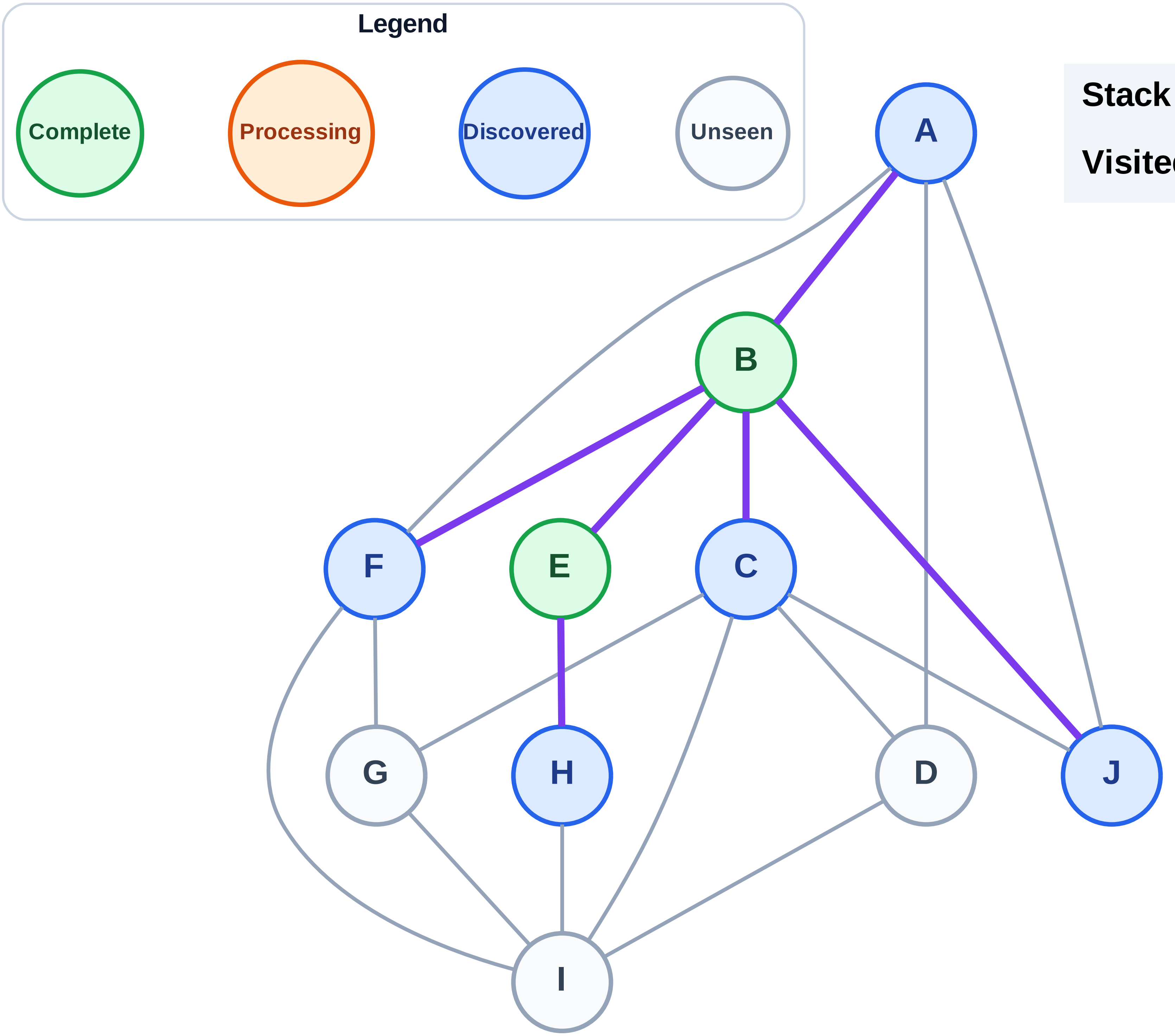
DFS Traversal

Step 4: Process node B



DFS Traversal

Step 5: Discover J, F, C, A from B

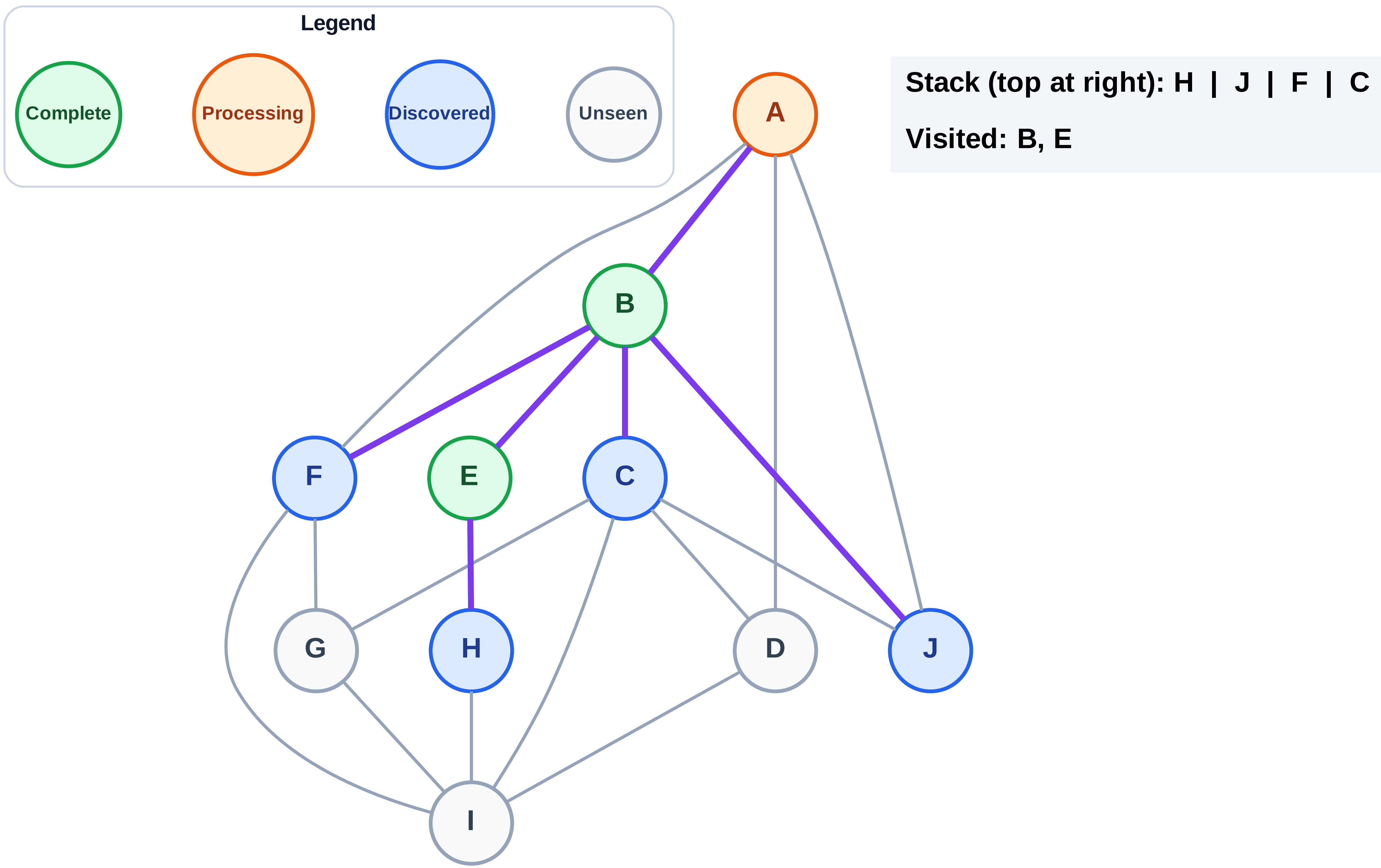


Stack (top at right): H | J | F | C | A

Visited: B, E

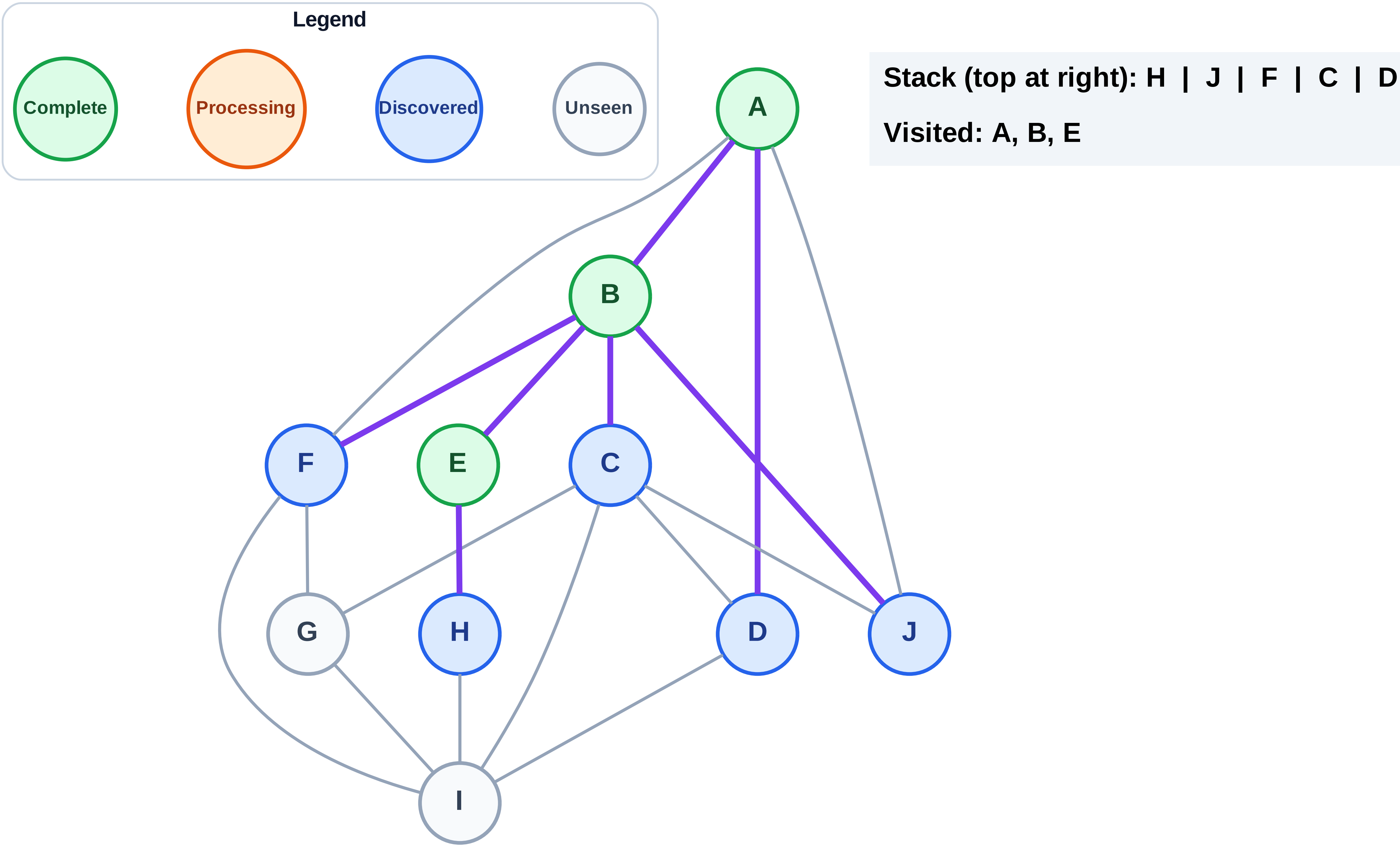
DFS Traversal

Step 6: Process node A



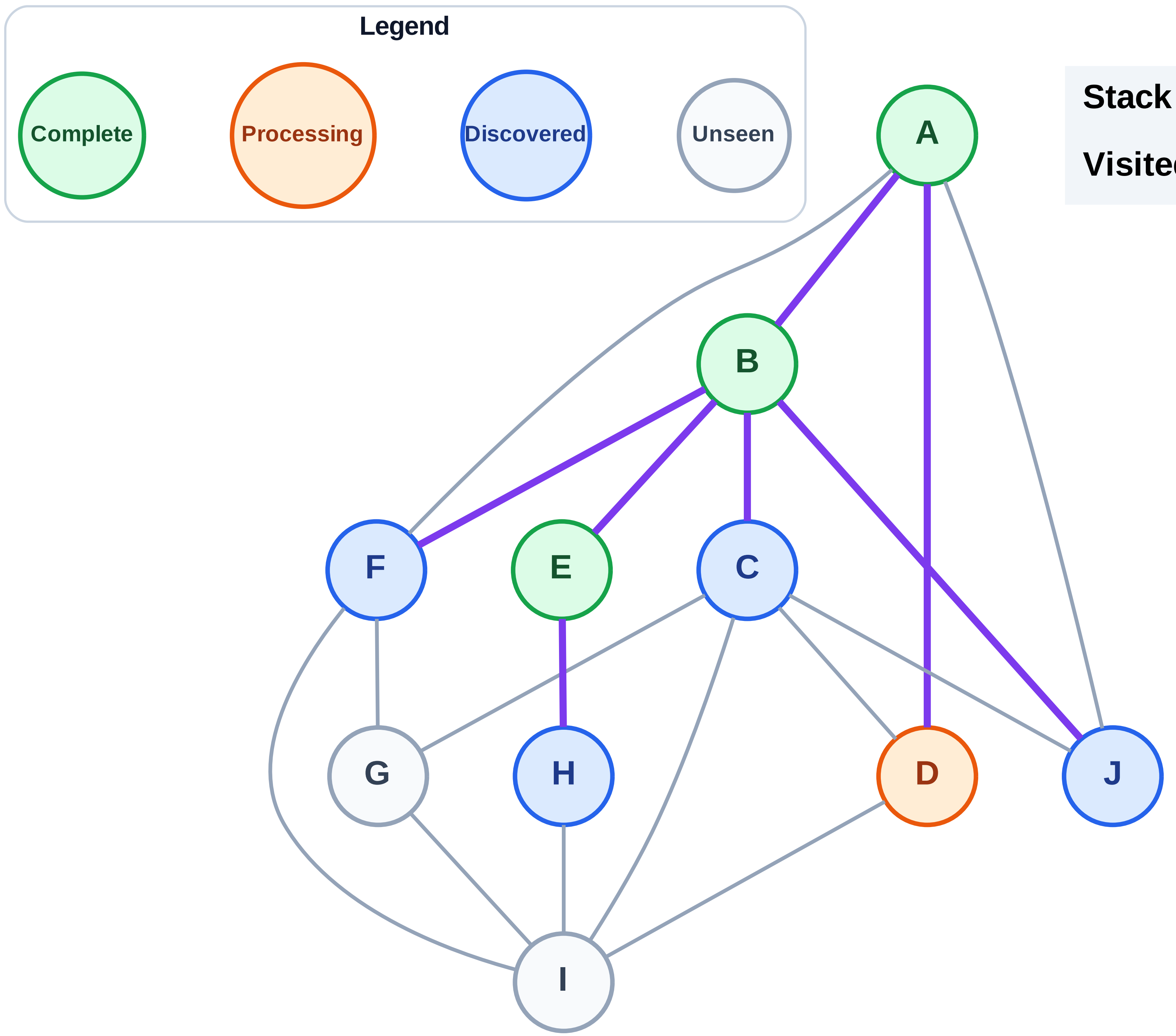
DFS Traversal

Step 7: Discover D from A



DFS Traversal

Step 8: Process node D

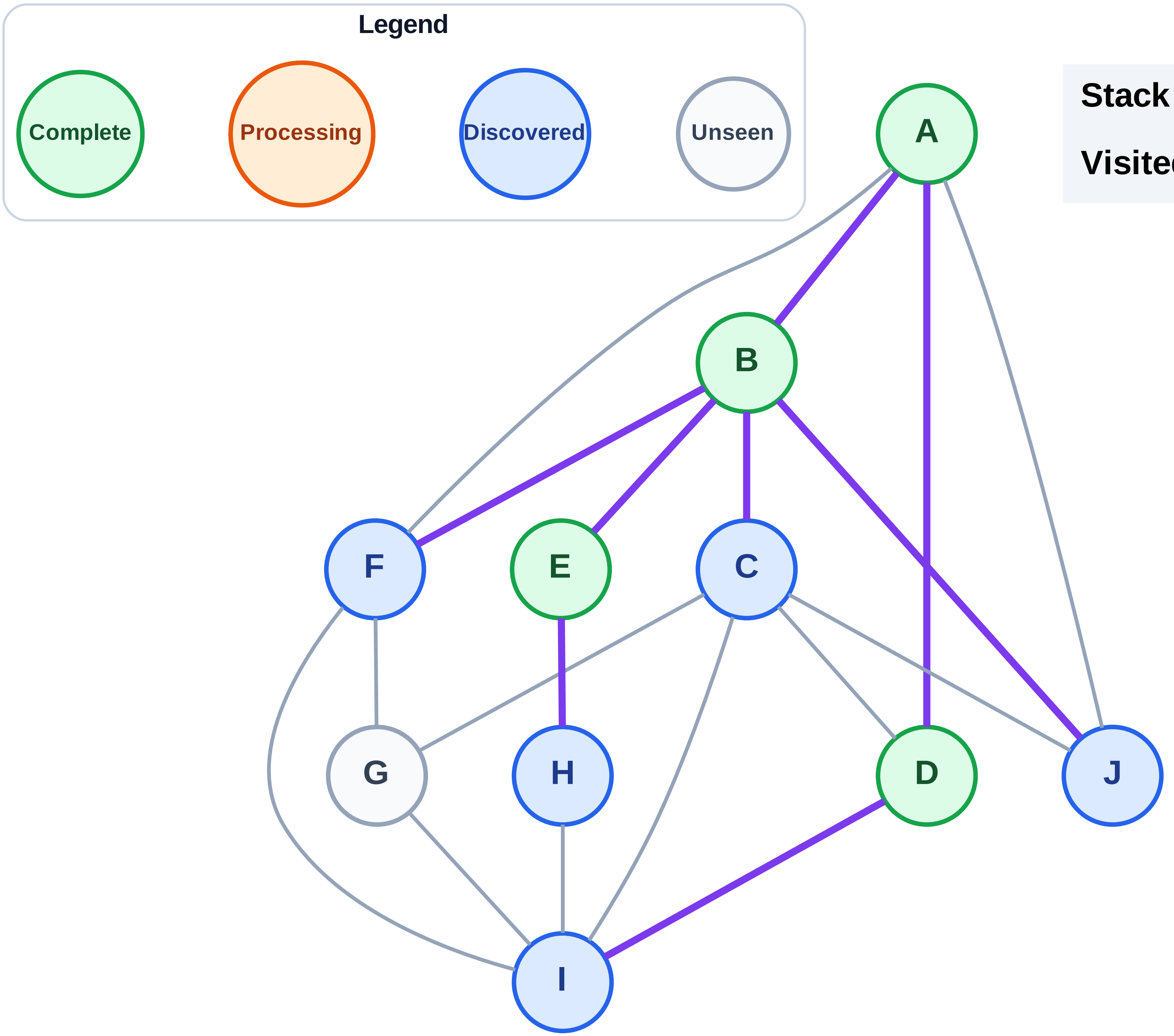


Stack (top at right): H | J | F | C

Visited: A, B, E

DFS Traversal

Step 9: Discover I from D

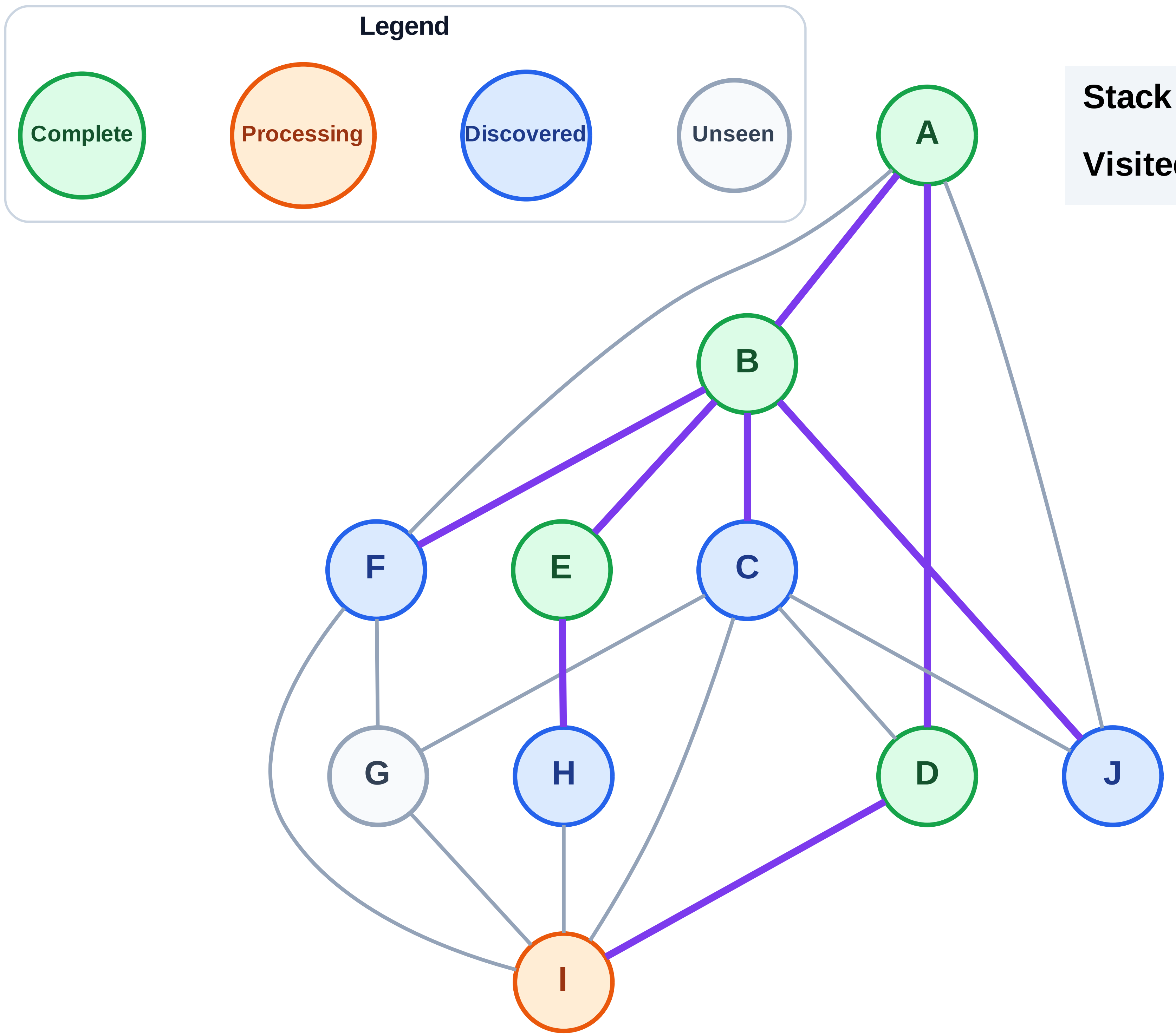


Stack (top at right): H | J | F | C | I

Visited: A, B, D, E

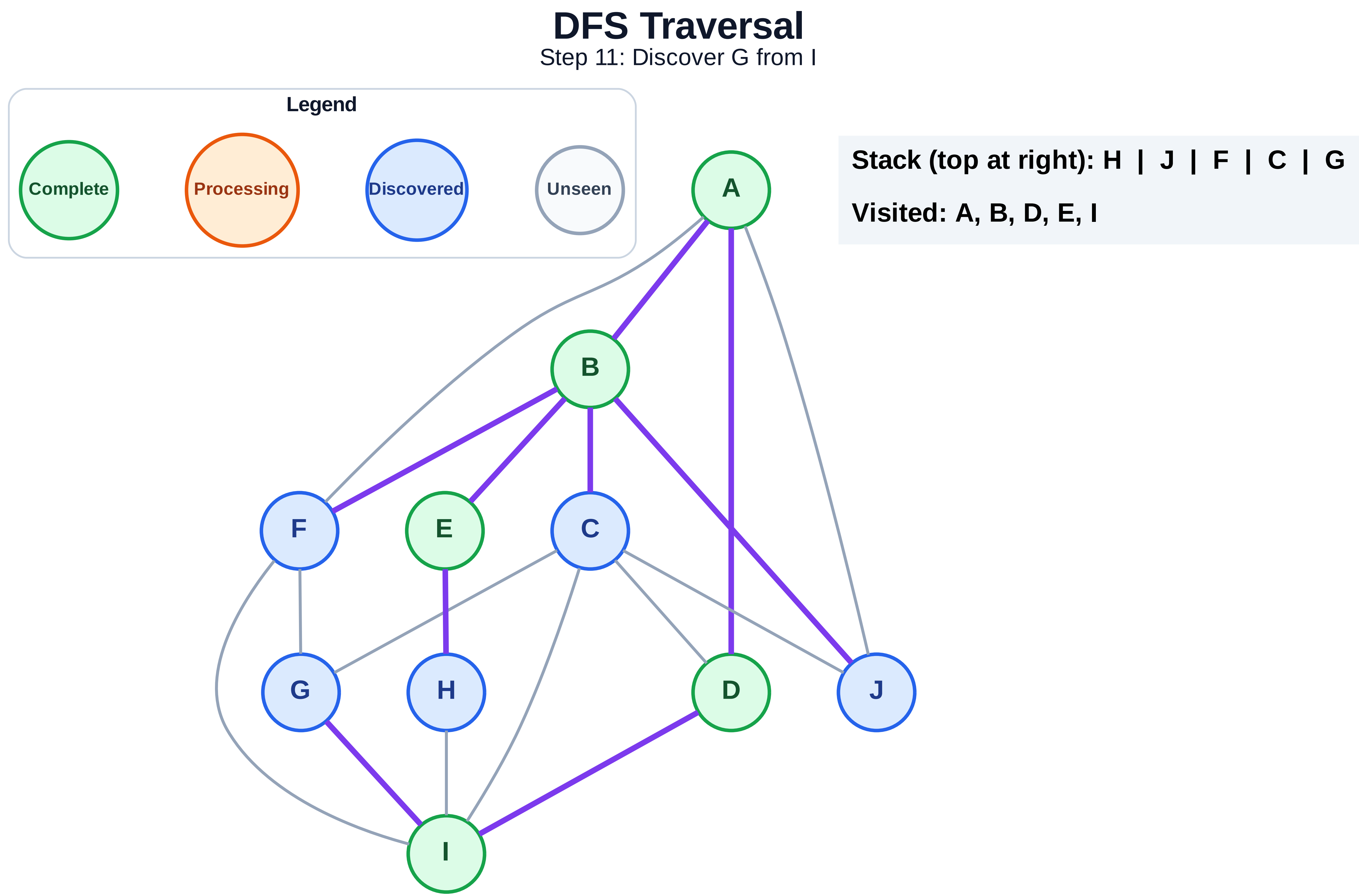
DFS Traversal

Step 10: Process node I



Stack (top at right): H | J | F | C

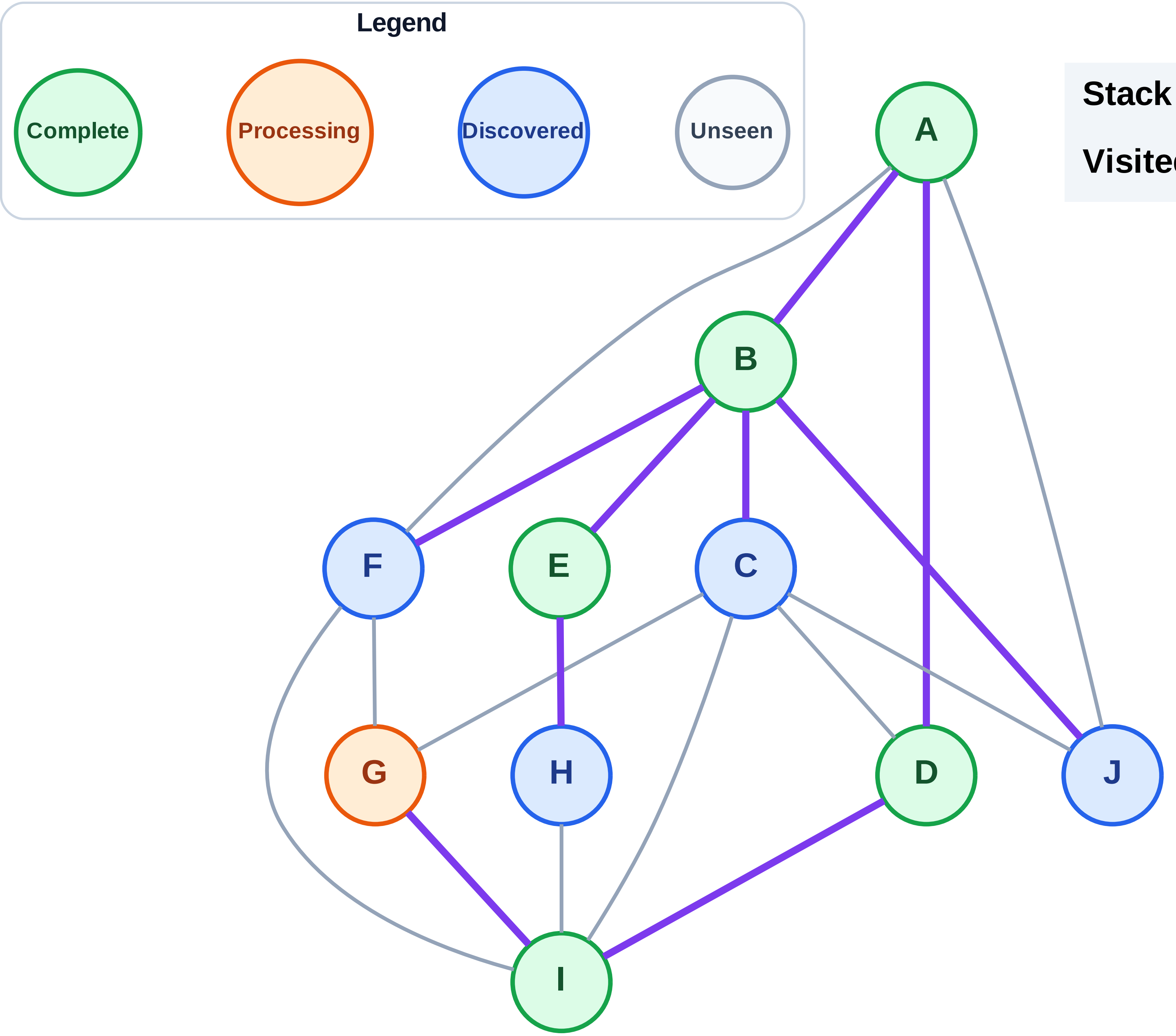
Visited: A, B, D, E



DFS Traversal
Step 11: Discover G from I
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): H | J | F | C | G
Visited: A, B, D, E, I
A
B
C
D
E
F
G
H
I
J

DFS Traversal

Step 12: Process node G

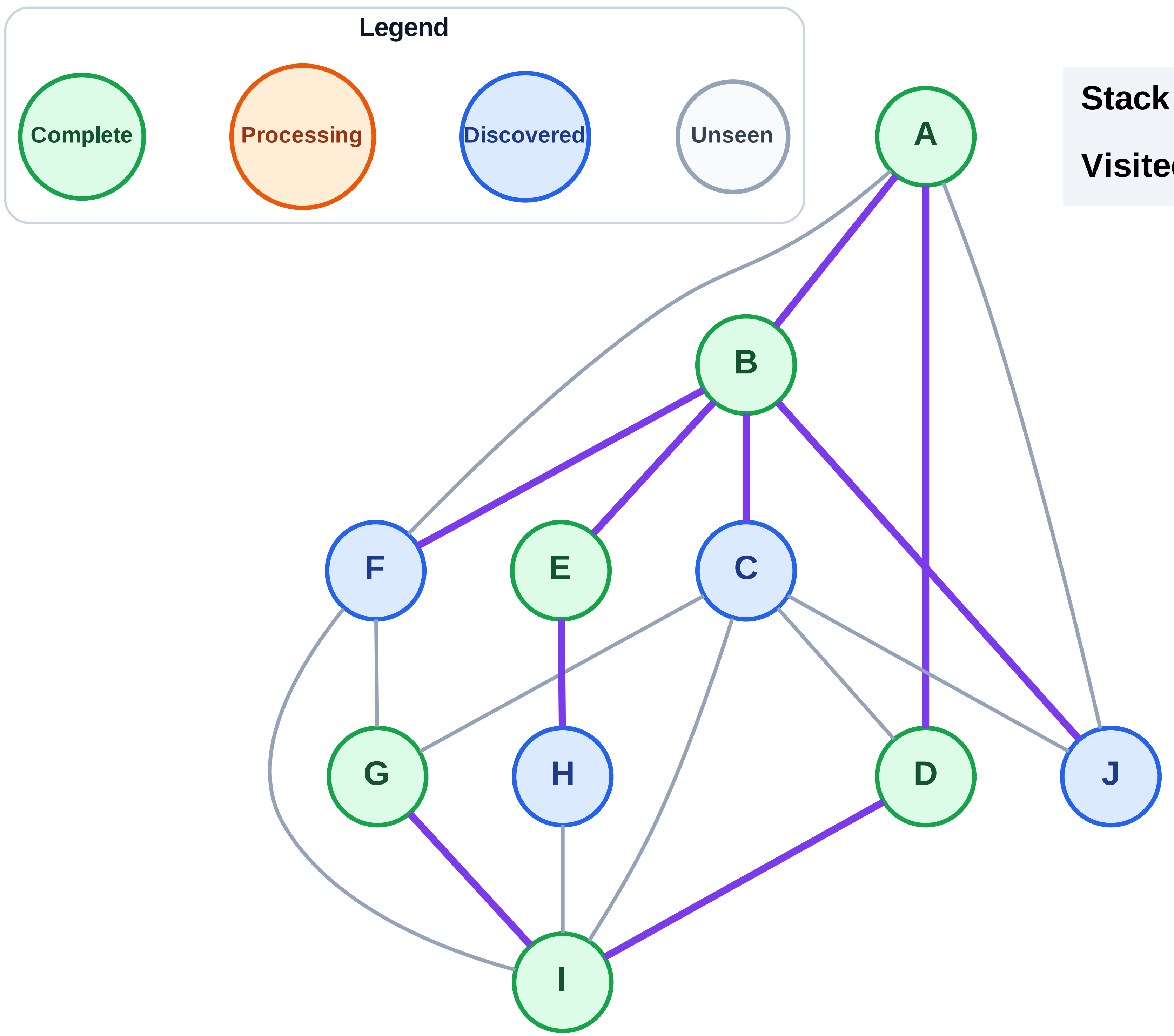


Stack (top at right): H | J | F | C

Visited: A, B, D, E, I

DFS Traversal

Step 13: No new neighbors from G

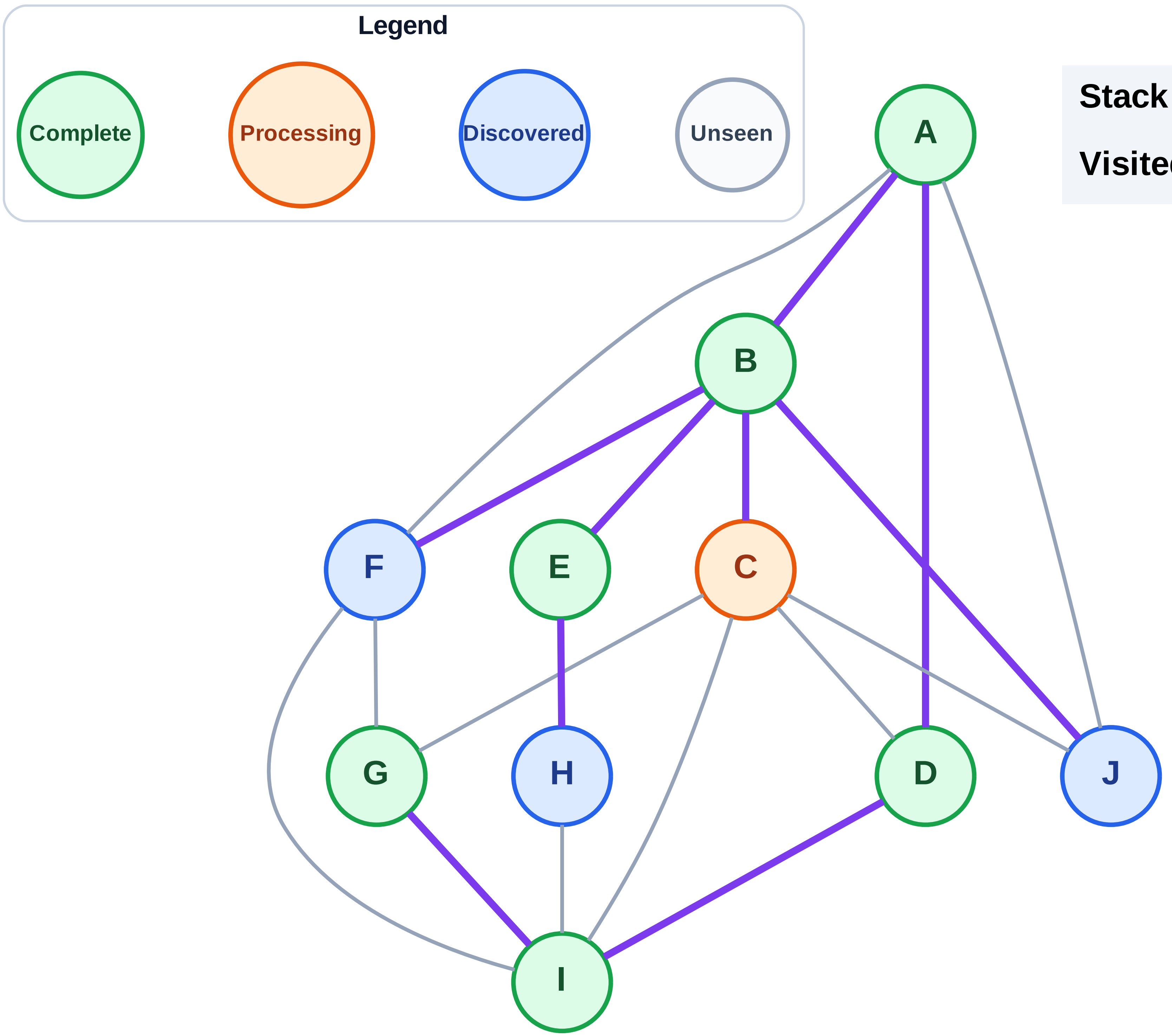


Stack (top at right): H | J | F | C

Visited: A, B, D, E, G, I

DFS Traversal

Step 14: Process node C

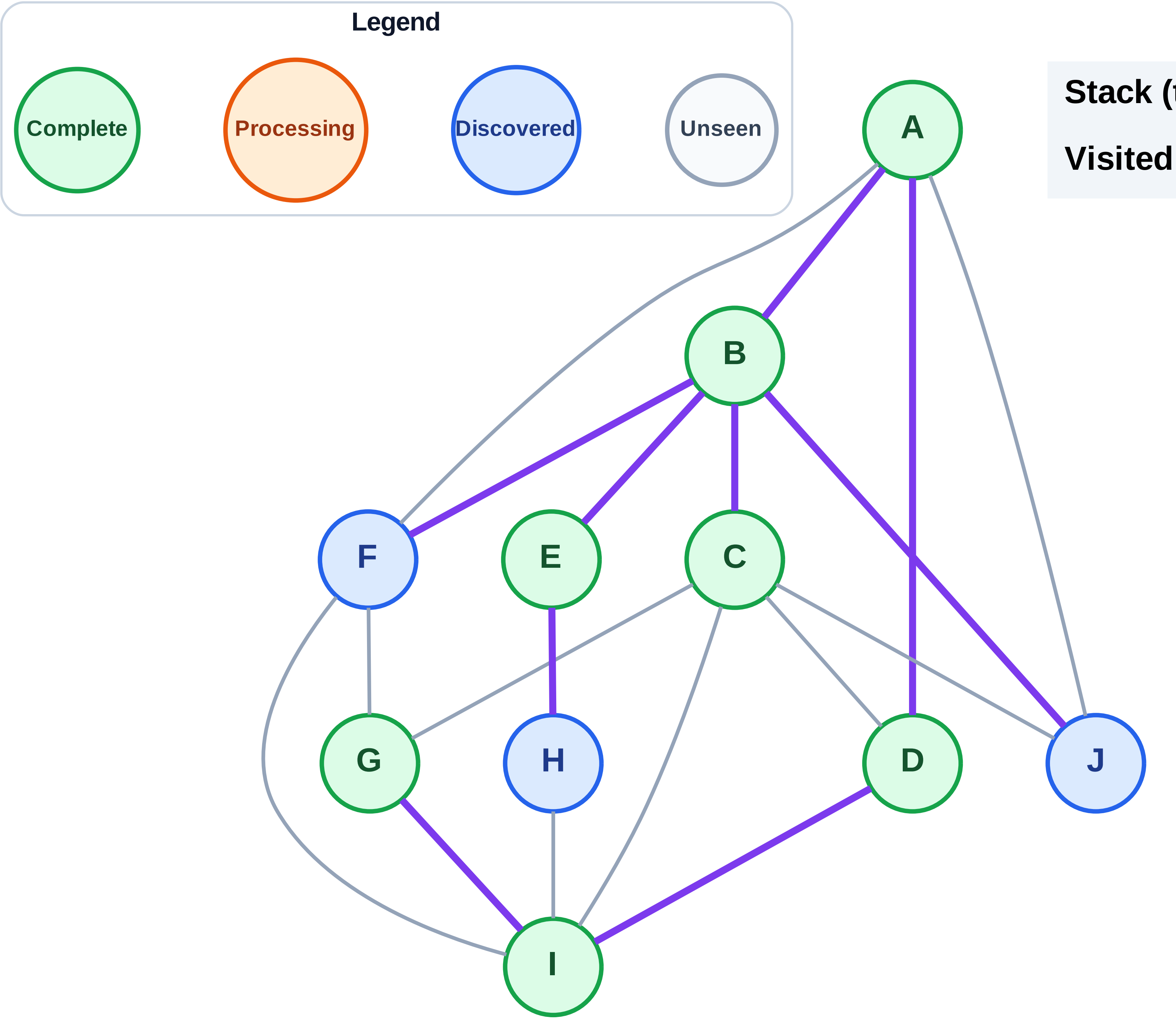


Stack (top at right): H | J | F

Visited: A, B, D, E, G, I

DFS Traversal

Step 15: No new neighbors from C

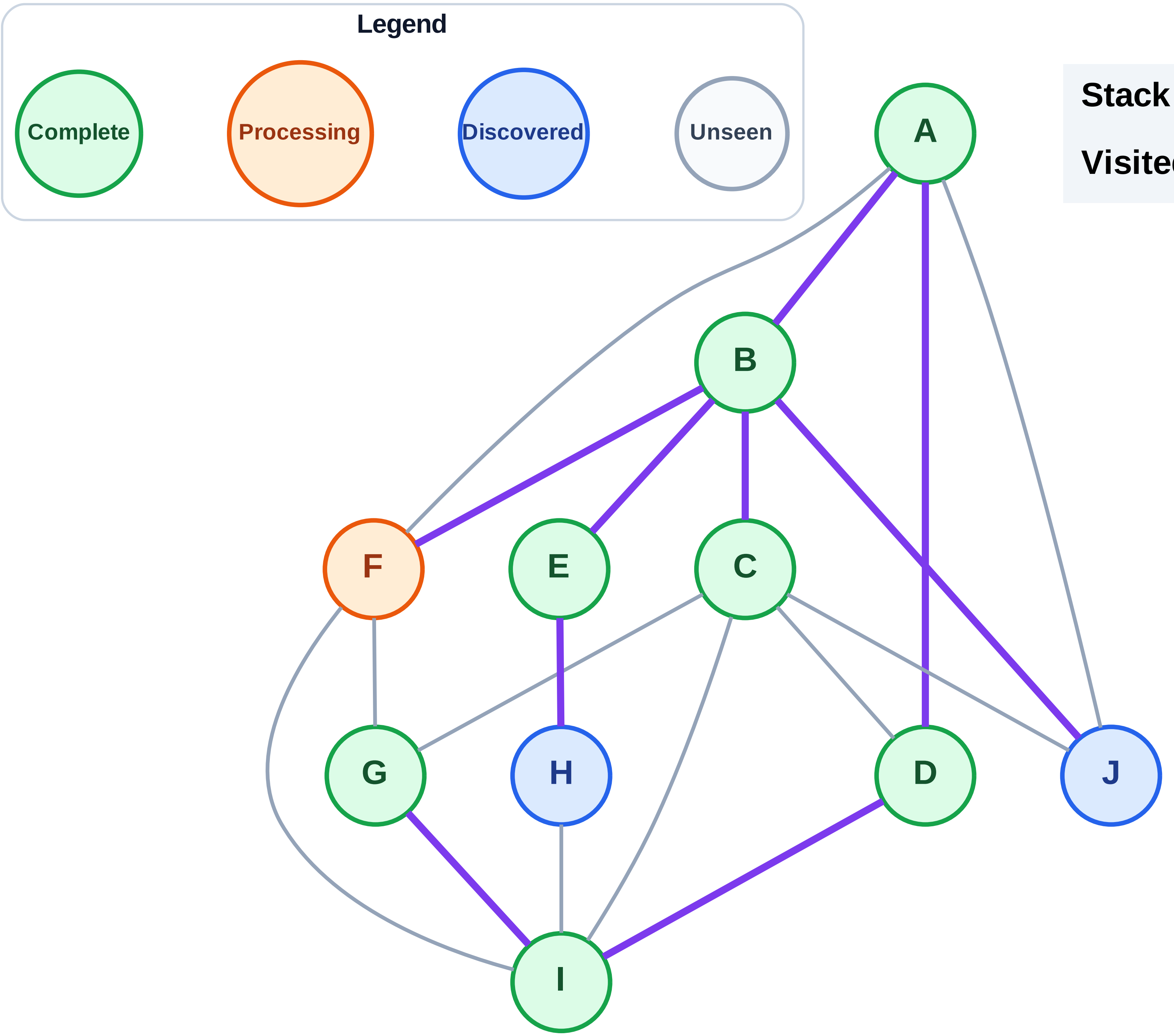


Stack (top at right): H | J | F

Visited: A, B, C, D, E, G, I

DFS Traversal

Step 16: Process node F

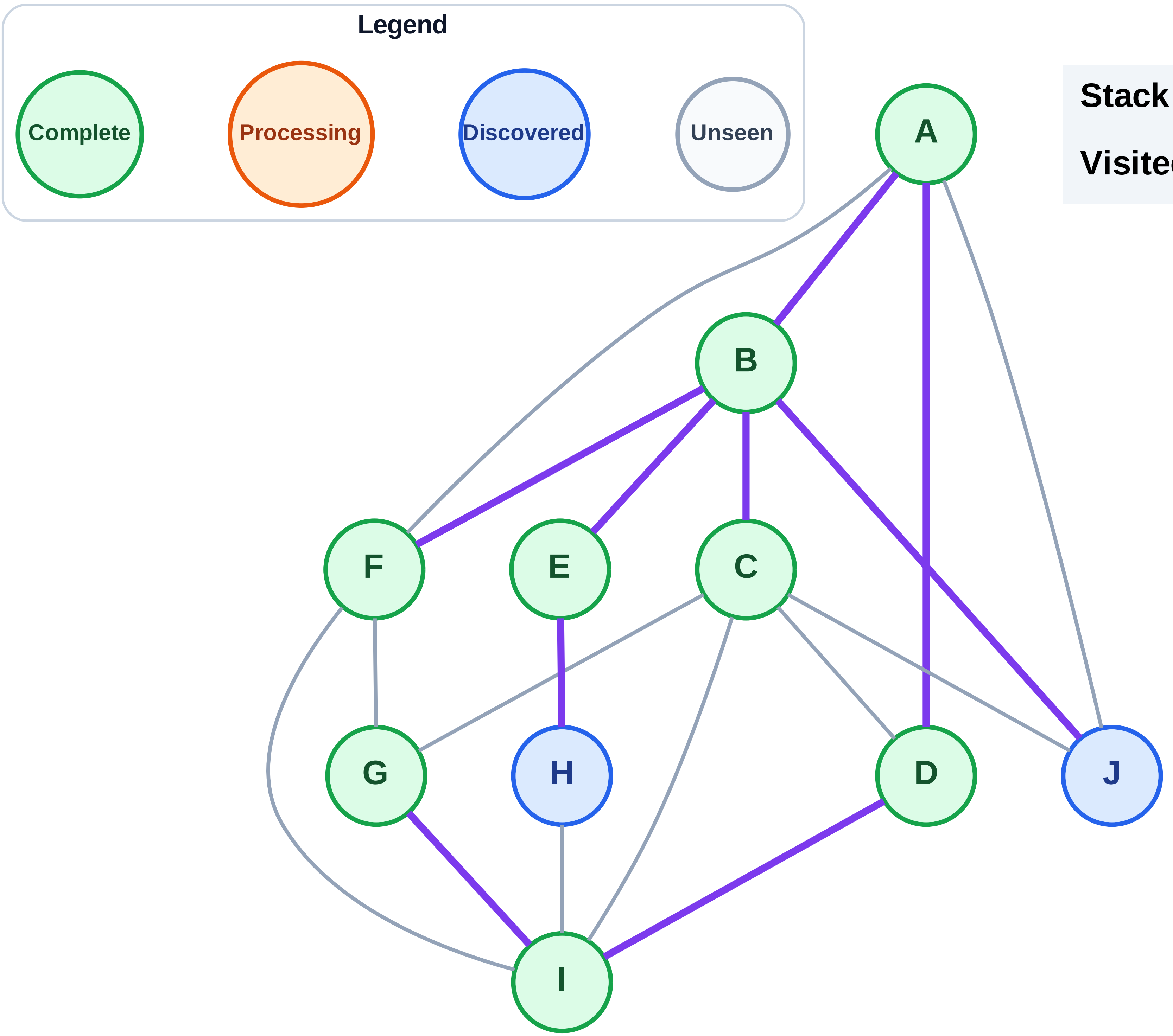


Stack (top at right): H | J

Visited: A, B, C, D, E, G, I

DFS Traversal

Step 17: No new neighbors from F

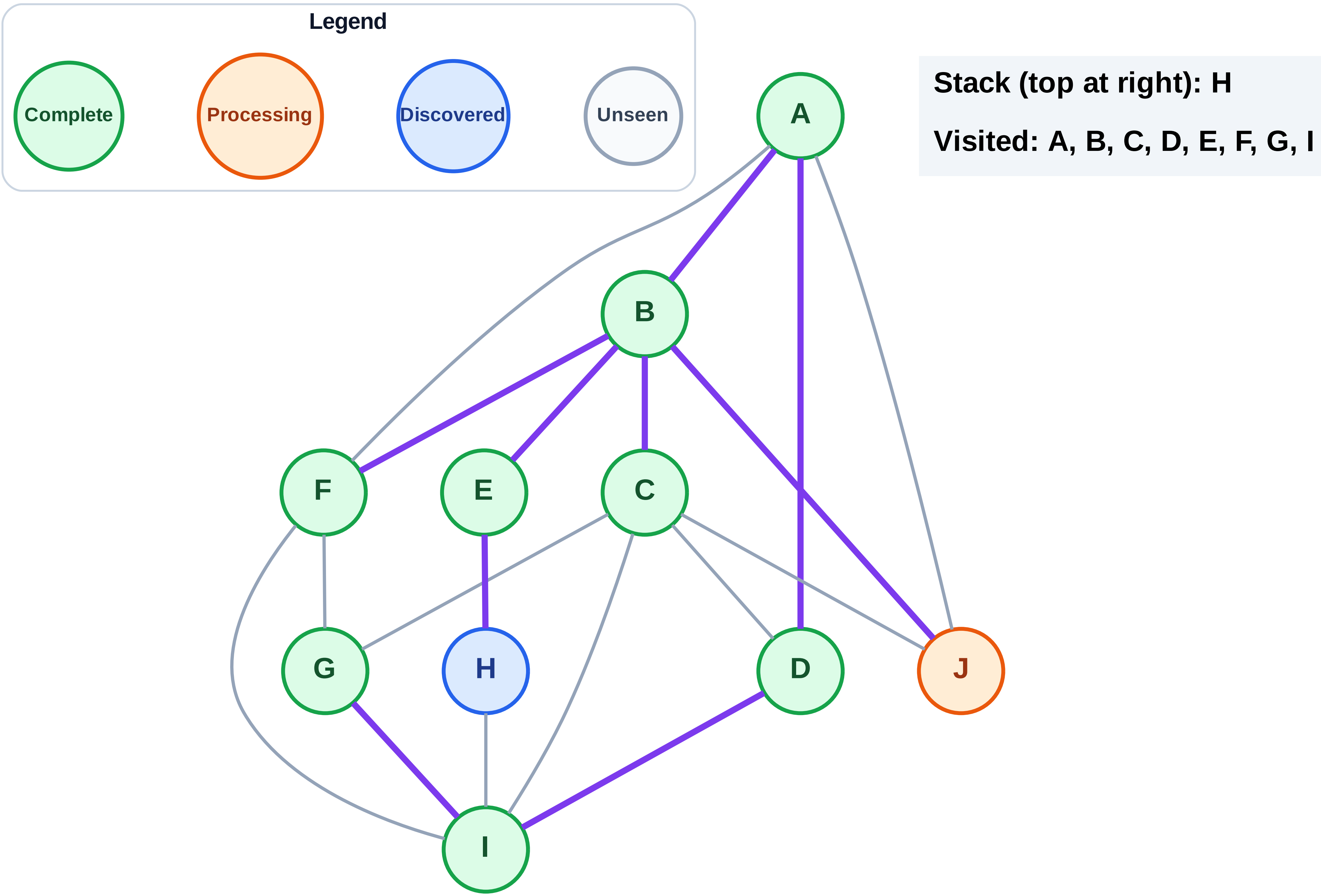


Stack (top at right): H | J

Visited: A, B, C, D, E, F, G, I

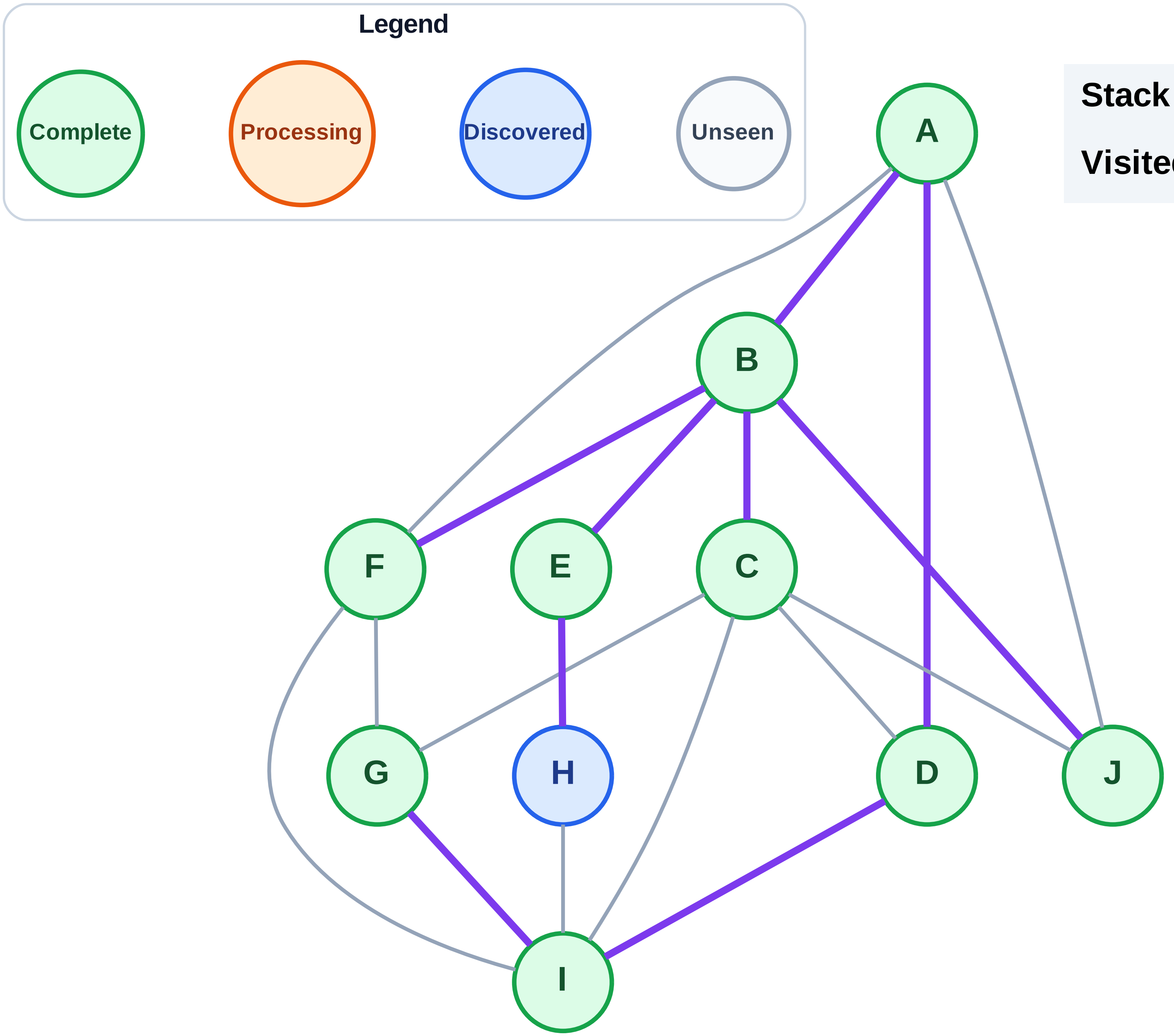
DFS Traversal

Step 18: Process node J



DFS Traversal

Step 19: No new neighbors from J

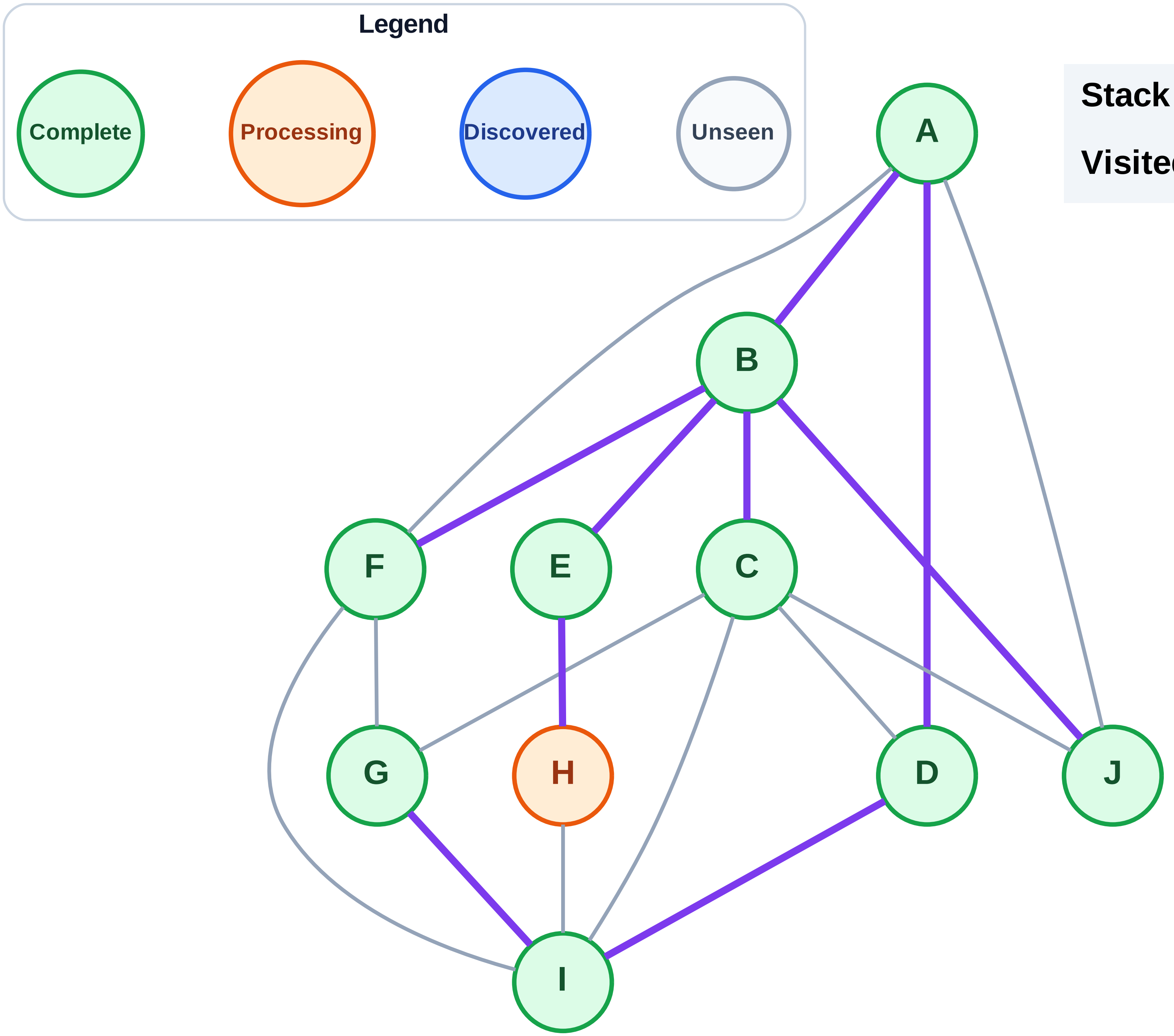


Stack (top at right): H

Visited: A, B, C, D, E, F, G, I, J

DFS Traversal

Step 20: Process node H

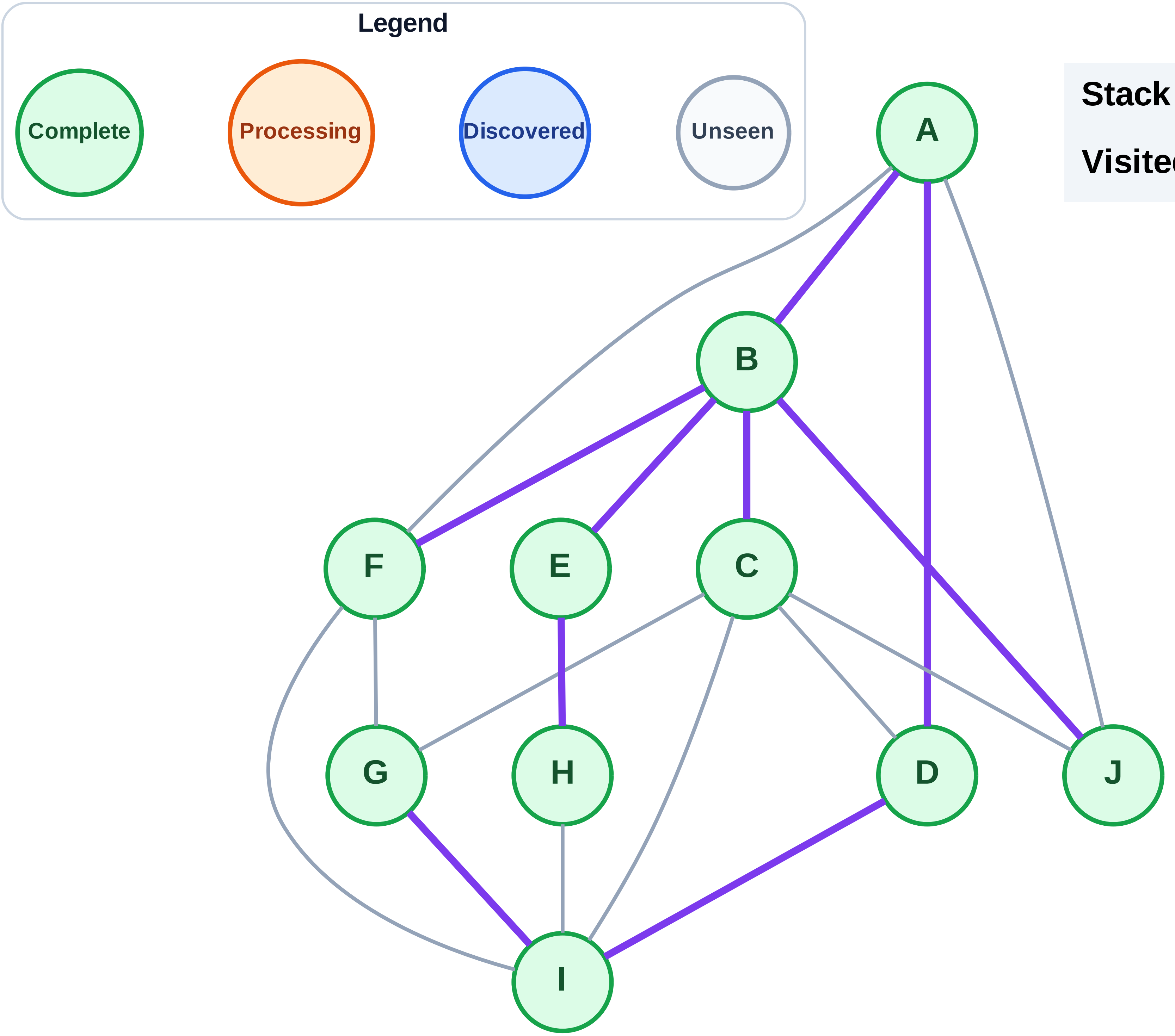


Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, I, J

DFS Traversal

Step 21: No new neighbors from H



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J